



Matheson Constructors Limited
245 Industrial Pkwy North
Aurora, Ontario L4G 4C4
P: (905) 669-7999
F: 905 669-0268

Project: 2025-048 Dockstader PRPS Reporting
Station
6 Dockstader Rd.
Brampton, Ontario L6R

Submittal #08 11 13-1.1 - SR-009 - Revised HM Doors and Frames - Metal Door Ltd.

08 11 13 - STEEL DOORS AND FRAMES

Revision	1	Submittal Manager	Alvin Hernandez (Matheson Constructors Limited)
Status	Open	Date Created	Jul 28, 2025
Issue Date	Jul 28, 2025	Spec Section	08 11 13 - STEEL DOORS AND FRAMES
Responsible Contractor	Egress Systems of Canada Ltd.	Received From	Scott Minhinnick (Egress Systems of Canada Ltd.)
Received Date		Submit By	May 23, 2025
Final Due Date	Oct 28, 2025	Lead Time	
		Cost Code	
Location		Type	Substitution Request
Submittal Package			
Approvers	Alvin Hernandez (Matheson Constructors Limited), Ashish Kumar (Diamond Schmitt Architects)		
Ball in Court	Alvin Hernandez (Matheson Constructors Limited)		
Distribution	Ali Haidar (Matheson Constructors Limited), Alvin Hernandez (Matheson Constructors Limited), Kathy Schaeffer (Matheson Constructors Limited), Lizet Pabon (Matheson Constructors Limited), Marwan Diab (Matheson Constructors Limited), Mitch Minhinnick (Egress Systems of Canada Ltd.), Anna Goddard (Egress Systems of Canada Ltd.)		
Description	Enclosed for your review and acceptance is Submittal 08 11 13-01-R1-SR-009-Revised HM Doors and Frames - Metal Door Ltd		
Schedule ID #			

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					
Scott Minhinnick	Jul 28, 2025	Oct 17, 2025	Oct 17, 2025	Submitted	Submittal 08 11 13-01-R1-SR-009-Revised HM Doors and Frames - Metal Door Ltd.pdf
Alvin Hernandez	Oct 17, 2025	Oct 17, 2025	Oct 17, 2025	Reviewed as noted	
Ashish Kumar	Oct 17, 2025	Oct 28, 2025	Dec 16, 2025	Reviewed as noted	072R1 08 11 13-1.1 SR-009 - Revised HM Doors and Frames - Metal Door Ltd_DSA.pdf (Current) Transmittal_Submittal - 072R1 (Responded and Closed).pdf (Current)
Comment	See attached PDF				

NAME	COMPANY	EMAIL	PHONE
Ashish Kumar	Diamond Schmitt Architects	akumar@dsai.ca	1-416-862-8800

NAME	COMPANY	EMAIL	PHONE
Alvin Hernandez	Matheson Constructors	ashernandez@mathesonconstructors.com	905 669 7999
Ali Haidar	Matheson Constructors	ahaidar@mathesonconstructors.com	
Arne Suraga	Diamond Schmitt Architects	ASuraga@dsai.ca	1 (416) 862-8800 x516
Ashish Kumar	Diamond Schmitt Architects	akumar@dsai.ca	1-416-862-8800
Fariz Dhalla		fariz.dhalla@entuitive.com	
Hany Rezk	Diamond Schmitt Architects	hrezk@dsai.ca	1-416-862-8800
Haril Thakore		hthakore@cdml.ca	
Helal Zabardast	Diamond Schmitt Architects	hzabardast@dsai.ca	1-416-862-8800
Jamie Carmichael	CDML	jcarmichael@cdml.ca	1 (416) 583-2108
Kathy Schaeffer	Matheson Constructors	kschaeffer@mathesonconstructors.com	905 669 7999
Lizet Pabon	Matheson Constructors	lpabon@mathesonconstructors.com	905 669 7999
Marwan Diab	Matheson Constructors	mdiab@mathesonconstructors.com	905 669 7999
Tristan Crawford	Diamond Schmitt Architects	TCrawford@dsai.ca	1-416-862-8800 ext. 641

Submittal Transmittal

DATE: 12/16/2025
ID: 01175

REMARKS: See attached PDF

CONTENTS

QUANTITY: 1 DATED: 12/16/2025 NUMBER:

DESCRIPTION:

072R1 08 11 13-1.1 SR-009 - Revised HM Doors and Frames - Metal Door Ltd_DSA.pdf

ACTION:

REMARKS:

COPIES:

Ajit Stephen
Majadeh Hamidsaeidi

(Region of Peel Deveolpment Services - Planning)
(Region of Peel Deveolpment Services - Planning)

384 Adelaide Street West, Suite 100
Toronto, ON M5V 1R7
t: 416 862 8800

1050 West Pender Street, Suite 2010
Vancouver, BC V6E 3S7
t: 604 674 0866

1776 Broadway, Suite 2200
New York, NY 10019
t: 212 710 4329

www.dsai.ca
info@dsai.ca

To:	Matheson Constructors 245 Industrial Parkway North Aurora, ON L4C 4C4	Submittal No:	072R2
		Project No:	221033
		File No:	4-6-1
Attention:	Kathy Schaeffer	Date:	December-16-25
Project:	Dockstader PRPS		

The Architect's review is for the sole purpose of ascertaining conformance with the general design concept and for general arrangement. This review shall not mean approval of the detail design inherent in the shop drawings, responsibility for which shall remain with the Contractor and such review shall not relieve the Contractor of his responsibility for errors or omissions in the shop drawings or of his responsibility for meeting all requirements of the Contract Documents. The Contractor is responsible for all dimensions to be confirmed and correlated at the job site, for information that pertains solely to the fabrication processes, quantities or to techniques of construction and installation and for co-ordination with related work.

Contractor Package #	Spec Section	Description	Reviewed by	Status
08 11 13-1.1	08 11 13	SR-009 - Revised HM Doors and Frames - Metal Door Ltd.	DSA	RN

Status Legend: **R** – Reviewed **RN** – Reviewed As Noted **RR** – Revise and Resubmit **N** – Not Reviewed

Comments: Please see attached for comments.

Per: Hany Rezk

Submittal #08 11 13-1.1 - SR-009 - Revised HM Doors and Frames - Metal Door Ltd.

08 11 13 - STEEL DOORS AND FRAMES

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Alvin Hernandez	Oct 17, 2025	Oct 17, 2025	Oct 17, 2025	Reviewed as noted	
Ashish Kumar	Oct 17, 2025	Oct 28, 2025		Pending	

DSA-HR

Substitution request for HM doors and frames is acceptable subject to comply with all contract documents requirements
- Provide all required submittals, shop drawings, door schedules and data sheets for consultants reviews

245 Industrial Parkway North
Aurora, ON L4G 4C4
Phone: (905) 669-7999
Fax: 905-669-0268

Date: October 17, 2025 Job No.: 2025-048

To: Diamond Schmitt

Attn: Laura Hutchinson / Ashish Kumal

C.C.: Project File, A. Stephens, K. Schaeffer, M. Diab, A. Hernandez

Re: Submittal 08 11 13-01-R1-SR-009-Revised HM Doors and Frames - Metal Door Ltd

WE ARE SENDING YOU 1 electronic copy via email of the following items:

ZERO CARBON DOCUMENTS	SHOP DRAWINGS	PRODUCT INFORMATION	SAMPLES	SUBSTITUTION REQUEST			
				X			

NO.	DATE	REV #	DESCRIPTION
1	Oct. 17/25	1	Completed Revised Substitution Request Form: HM Doors and Frames – Metal Door Ltd.
2	Oct. 17/25	0	Metal Door Ltd. email regarding CSDMA and NAAMM dated Aug. 14/25
3	Oct. 17/25	0	List of Projects
4	Oct. 17/25	0	List of Projects completed with DSA.
5	Oct. 17/25	1	Door and Frame Shop Drawings
6	Oct. 17/25	0	Storage And Installation of Hollow Metal Doors and Frames

THESE ARE TRANSMITTED as checked below:

☒ For approval

☐ For signature

☐ For your information

☐ As requested

☐ For review and comment

REMARKS:

Alvin Hernandez

Signature

This transmission, including any accompanying attachments, contains information that may be privileged and/or exempt from disclosure under applicable law, and is intended only for the use of the recipient(s) named above. Any distribution, review, dissemination, or copying of the contents of this communication by anyone other than the intended recipient(s) is strictly prohibited. If you receive this communication in error, please notify the sender immediately and destroy the copy you have received. Thank you.

1. GENERAL

1.1 Project Information

1.1.1

1.1.2 Project: DOCKSTEADER PRPS

1.1.3 From: Egress Systems of Canada Ltd

1.1.4 Date: ~~June 25, 2025~~ Updated REV02 September 26, 2025

1.2 Proposed Substitution

REFERENCE INFORMATION

Specification Section Number and Title: Section 08 11 13 Steel Doors and Frames

Drawing Reference (if applicable):

Page: Page 6, 2.1 Manufacturers

Article/Paragraph:
2.1.1

Specified Product: Steel Doors and Frames

SUBSTITUTION INFORMATION

Proposed Substitution (Product Name): Metal Door Ltd, Ontario, Canada

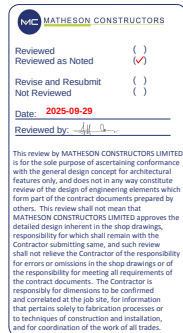
Product Description: Steel Doors and Frames

Product History: ☐ New Product ☐ 2-5 years old ☐ 5-10 years old ☒ > 10 years old

Manufacturer: Metal Door Ltd

Standard Warranty Offered: Yes Extended Warranty Available? ☒
Yes ☒ No

Address: 6 Stalwart - Industrial Drive, Gormley, ON L0H 1G0



Name of Technical Representative (TR): Dave Howe

Technical Representative's Contact

Phone 1-905-888-5151

Email: dhowe@metaldoorltd.com

Trade Name: Egress Systems of Canada Ltd (Distributor) via Metal Door Ltd (Manufacturer)

Differences between proposed substitution and specified product:

Identical

Reason for not providing specified item:

Requirement detailed manufacturers that could not comply and/or did not bid the project. The specification notes that 1.8.1 Manufacturers: Execute work in this Section by a manufacturer who is a member of CSDMA. Metal Door Ltd is a member that could comply with specified requirements.

Will proposed substitution affect other parts of Work? ☒ No ☐ Yes;
explain _____

Cost Implications of Proposed Substitutions: ☒ None ☐ Savings/Credit; ☐ Extra Costs
(explain below)

Embodied Carbon Impact (GWP) of Proposed Substitutions: ☒ None ☐ Reduced; ☐
Increase (explain below)

REFERENCE INSTALLATIONS

Consultant: The projects noted here are currently under construction. However, please refer to Metal Door Ltd letter dated September 25, 2025 for a list of previous projects completed with DSA

List below installations that reference the product installed in projects that are similar in scope and size. Provide at least 3 pertinent reference installations

PROJECT NO.1

Project Title: Peel Regional Police

Project Description: ☒ Commercial ☐ Residential ☐ Retail ☐ Healthcare ☐ Other:

Date Installed: Underway Location: Peel

Reference Contact (if known): Phone: Eastern Construction Email:

PROJECT NO.2

Project Title: Clarington Police

Project Description: ☒ Commercial ☐ Residential ☐ Retail ☐ Healthcare ☐ Other:

Date Installed: Underway Location: Peel

Reference Contact (if known): Phone: _____ Email: _____
Buttcon Construction

PROJECT NO.3

Project Title: Toronto Police Services

Project Description: ☒ Commercial ☐ Residential ☐ Retail ☐ Healthcare ☐ Other:

Date Installed: Underway Location: _____

Reference Contact (if known): Phone: _____ Email: _____
Eastern Construction

SUSTAINABLE INFORMATION

Indicate below sustainable information that apply to this Product.

MANUFACTURING LOCATION

What is the location of the manufacturing plant for this particular product?
6 Stalwart - Industrial Drive, Gormley, ON L0H 1G0

ENVIRONMENTAL PRODUCT DECLARATION (EPD)

Does the product have a compliant environmental product declaration (EPD)? ☐ Yes
☒ No

EPD Type: ☐ Industry Wide EPD ☐ Product Specific EPD

Indicate EPD expiry date: N/A

MATERIALS AND RESOURCES

Does the product contain wood Products? ☐ Yes ☒ No

If yes, Indicate percentage of FSC Certified wood in product:

Indicate wood product low-emitting characteristics:

☐ No Added Urea Formaldehyde (NAUF) ☐ Ultra-low Emitting Formaldehyde (ULEF)

Does the product contain post-consumer or pre-consumer recycled content? ☐ Yes ☐ No

If yes, Indicate percentage of pre-consumer recycled content:

If yes, Indicate percentage of post-consumer recycled content:

DISCLOSURE AND TRANSPARENCY

Does the product participate in extended producer responsibility program? ☒ Yes ☐ No

Does the product have a "Declare" label with ingredient disclosure greater than 1000 ppm? ☐ Yes ☒ No

Does the product have a Fully Declare Health Product Declaration? ☐ Yes ☒ No

Is this product CDPH Emissions testing compliant? ☐ Yes ☒ No

Indicate product VOC content (g/l):

SUPPORTING DATA

Attached data includes product description, specifications, drawings, photographs, and performance and test data adequate for evaluation of the request; applicable portions of the data are clearly identified.

Attached data also includes a description of changes to the Contract Documents that the proposed substitution will require for its proper installation

Supporting Data Attached: ☐ Drawings ☐ Product Data ☐ Samples ☐ Test Reports <https://metaldoorltd.com>
<https://csdma.org/member-directory>

FOR INTERNAL USE ONLY

Substitution approved - Make submittals in accordance with Section 01 25 00.
Substitution approved as noted - Make submittals in accordance with Section 01 25 00.

Substitution rejected - Use specified materials.

1.3 DECLARATIONS

- 1.3.1 I/We the undersigned agree that the Owner reserves the right to accept or reject any or all the proposed substitution/alternatives and may request that materials specified in the Bidding and Contracting Documents be used.
- 1.3.2 I/We the undersigned understand that by submitting this Substitution Request, I/We assume full responsibility for ensuring that all requirements are considered.
- 1.3.3 I/We hereby certify that:
- .1 Proposed substitution has been fully investigated and determined to be equal or superior in all respects to specified product.
 - .2 Same warranty will be supplied for proposed substitution as for specified product.
 - .3 Same maintenance service and source of replacement parts, as applicable, is available.
 - .4 Proposed substitution will have no adverse effect on other trades and will not affect or delay progress schedule.
 - .5 Proposed substitution does not affect dimensions and functional clearances.
 - .1 Coordination, installation, and changes in the Work as necessary for accepted substitution will be complete in all respects.

SUBMITTED BY:

Scott Minhinnick

Signature of

Duly Authorized

Signing Officer

Scott Minhinnick

SM

Affix Seal Here

Position: President

Firm: Egress Systems of Canada Ltd

Street Address: 243 Saunders Rd,

City, Province, Postal
Code Barrie, Ontario, L4N 9A3

Phone: 416-434-2100

Updated REV02 26

Updated September

Respectfully submitted
this

~~xxx~~
~~25~~ day of ~~June~~, 20~~25~~ at
This City of Barrie.

END OF FORM



METAL DOOR LTD.

Manufacturers of Steel Doors and Frames

Re: Dockstader PRPS

August 14/2025

Att:Scott

We are members in good standing with CSDMA and in the process of becoming members with NAAMM.

We have been manufacturing hm frames and doors for over 40 years for all different types of commercial projects. (see attached abbreviated project list).

Should you have any other questions or concerns, please feel free to contact me at your convenience.

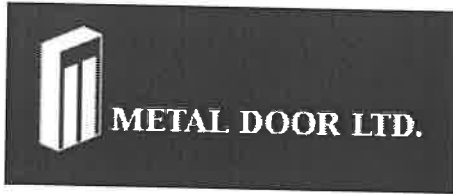
Regards,

Dave Howe/owner
Metal Door Ltd

P.O. Box 638 Gormley, Ontario L0H 1G0 Ph (905)888-5151 Fax (905)888-0647
Family owned and operated since 1980.
www.metaldoorltd.com



905-888-5151



Projects

HOSPITALS:

- Women's College Hospital – Toronto
- St Michaels Hospital – Toronto
- Georgian Bay Hospital – Midland
- Michael Garron Hospital – East York
- Sioux Lookout Health Centre – Sioux Lookout
- Kingston General Hospital – Kingston
- St. Mary's Hospital – Kitchener
- Hawkesbury Hospital – Hawkesbury
- St Joseph Care – Thunder Bay
- Princess Margaret Hospital – Toronto
- Royal Victoria Hospital – Cardiac – Barrie
- Sunnybrook Hospital – Toronto

SCHOOLS:

- Georgian College – Barrie
- Ontario Vet College – University of Guelph – Guelph
- Glen Campus School – Hamilton
- Markham District High School – Markham
- Stouffville District High School – Stouffville
- Bill Crothers Secondary School – Markham
- University of Western Science Building – London
- Fort Erie HS – Fort Erie
- University of Waterloo – Waterloo

- Kingston Secondary School – Kingston
- Memorial Elementary School – Hamilton
- McMaster Student Fitness – Hamilton
- Collegiate Ave Elementary School – Hamilton

OTHER INSTITUTIONS & FACILITIES:

- York Regional Police Headquarters – Newmarket
- CTR Excellence Transportation & Garage – Toronto
- East Gwillimbury Operations Centre – East Gwillimbury
- Hydro One – Orillia & Ottawa
- Milton RCMP – Milton
- Orangeville OPP – Orangeville
- Peel Police – Brampton
- Union Station – Toronto
- Caledon Fire Station – Caledon
- CFB Borden – Barrie
- Whitchurch Stouffville Library – Stouffville
- Barrie Emergency Services Campus – Barrie
- Collins Bay Institution – Kingston
- Peel Regional Police – Brampton
- Amazon – London & Cambridge
- Magna Centre – Newmarket
- Angus Glen Community Centre – Markham
- Pan Am – Milton/ Hamilton/ York
- Aurora Community Centre – Aurora
- Central East Correctional Centre – Lindsay
- Kingston Penitentiary – Kingston

Our Customers

Updated REV02 - September 25, 2025

To whom it may concern:

Metal Door Ltd have worked together with Diamond Schmitt for over a decade.
We are featured on many of their past signature projects.

- Windsor Law
- Huttonville School
- York Woods Library
- University of Guelph
- Chris Gibson Recreation Ctr
- Photos of projects are available on the Diamond and Schmitt Website.

Metal Door Ltd. Also provided all hollow metal products for
Peel Manor Long Term Care. and Peel Region Paramedic Station
While teamed up with Egress Systems

John Zeiger

Recognized By



Canadian Steel Door
Manufacturers Association
L'Association Canadienne
Des Fabricateurs de Portes D'Acier



Canada



6 Stalwart Industrial Dr. Gormley, Ontario Canada

WWW.METALDOORLTD.COM

MC MATHESON CONSTRUCTORS	
Reviewed	(✓)
Reviewed as Noted	()
Revise and Resubmit	()
Not Reviewed	()
Date: 2025-09-29	
Reviewed by: 	
This review by MATHESON CONSTRUCTORS LIMITED is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of engineering elements which form part of the contract documents prepared by others. This review shall not mean that MATHESON CONSTRUCTORS LIMITED approves the detailed design inherent in the shop drawings, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the shop drawings or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.	

SECTION 08 11 13 SHOP DRAWING REV02

FOR

DOCKSTEADER PRPS

PEEL REGIONAL PARAMEDIC SERVICES
6 DOCKSTEADER ROAD, BRAMPTON, ON

ES25-015

VERSION DATE
11-Oct-2025 11:43 PM
SM25-023

GENERAL CONTRACTOR:

MATHESON CONSTRUCTORS
245 INDUSTRIAL PARKWAY NORTH
AURORA, ON. L4G 4C4
T: (905) 669-7999 F: (905) 669-0268

ARCHITECT:

DIAMOND SCHMITT ARCHITECTS INC
384 ADELAIDE ST. W. SUITE 100
TORONTO, ON. M5V 1R7
T: (416) 862-8800 F: (416) 862-5508

CONSULTANT

Scott Minhinnick
scott@egress-systems.com

PROJECT MANAGER

Mitch Minhinnick
mitch@egress-systems.com

EGRESS

SYSTEMS OF CANADA LTD

243 Saunders Road
Barrie, Ontario. Canada
L4N 9A3

T: (705) 733-3667

www.egress-systems.com

PLEASE ADVISE ANY QUESTIONS OR NOTES ON THE SCHEDULE REMARKS COLUMN OR ELEVATION PAGES.



ALL DOORS, FRAMES AND SCREENS ARE MANUFACTURED AS PER SECTION IN SPECIFICATION 08 11 13

Frame Fire labels are combination labels suitable for 20 min / 3/4 / 1 hr / 1 1/2 hr.
3/4 hour Pemko Screen Labels required where glass exceeds testing dimension limits and or 2116 sq inches per lite.
SEE SCHEDULE REMARKS FOR ANY FIRE RATING QUIRIES OR LABEL RESTRICTION NOTES.
FIRE RATED SCREENS CAN NOT EXCEED 3910mm IN WIDTH AND 3250mm IN HEIGHT ARCHITECT WILL NEED TO ADVISE WHEN THIS APPLIES

MATERIALS

14 GA - 16 GA - 18 GA (ZF120) SEE CSDMA GAUGE CHART PROVIDED IN THE SHOP DRAWINGS
HOT DIP GALVANIZED (SATIN COAT / WIPE COAT) MEETING REQUIREMENTS OF ASTM A924 AND A653
HOT DIP GALVANIZED (WIPE COAT) MEETING REQUIREMENTS OF ASTM A924 AND A653
14 GA G90 - 16 GA G90 - 18 GA G90
Z275 / ZF275 (G90) HOT DIP GALVANIZED STEEL COMPLYING WITH ASTM A653/653M
ZINC PRIMER APPLIED AT ALL DOOR, FRAMES AND SCREENS AT ALL WELDED AND GRINDED AREAS

ABBREVIATIONS

W/C = WIPE COAT(SATIN COAT)
G90 = Z275 HOT DIP GALVANIZED
HM = HOLLOW METAL
WD = WOOD
BL = BORROWED LITE
PR = PAIR
LR = LEFT HAND AND RIGHT HAND
L = LEFT HAND
R = RIGHT HAND
DA = DOUBLE ACTING
T = THERMALLY BROKEN
HT = HIGH TRAFFIC
EXT = EXTERIOR
INT = INTERIOR

SEE DOOR AND FRAME TYPES
FOR ANY REMAINING ABBREVIATIONS

HOLLOW METAL TOLERANCES IS +/- 1.5 MM
ANY REQUIRED SITE MEASUREMENTS WILL BE
THE RESPONSIBILITY OF THE CONTRACTOR.

GLASS AND GLAZING NOTES.
ALL DOORS, FRAMES, AND SCREENS ARE MANUFACTURED AS PER SPECIFICATION DIVISION 8 (08 11 13)
INTERIOR AND EXTERIOR SCREENS OR TRANSOM FRAMES TO BE GLAZED WITH ADJUSTABLE GLASS STOPS UP TO 25mm
PENDING FIRE RATINGS AND MAXIMUM GLASS SPACE AND THICKNESS REQUIREMENTS.
ALL SCREWS FOR THE GLAZING COME COMPLETE WITH HOLLOW METAL PRODUCTS.
ALL INTERIOR GLASS TO BE SUPPLIED BY DIVISION 8 (08 81 13) PLEASE FOLLOW METAL DOOR SHOP DRAWINGS
ALL EXTERIOR GLASS TO BE SUPPLIED BY DIVISION 8 (08 81 23)
PLEASE FOLLOW METAL DOOR SHOP DRAWINGS FOR GLASS SIZE THICKNESS AND LOCATION OF GLASS SIDE.
METAL DOOR LTD. IS NOT RESPONSIBLE FOR INCORRECT GLASS INSTALLATION, LOST SCREWS OR GLAZING STOPS.
Glazing: Comply with requirements in Section 08 80 05, General Requirements for Glass and Glazing.

- 1. Exterior Glazing complete by Section 08 81 23 Exterior Glass and Glazing
- 2. Interior Glazing complete by Section 08 81 26 Interior Glass and Glazing

Recognized By



Dockstader PRPS

6 Dockstader Road, Brampton, ON
221033
File No. SPA-2024-0088

ARCHITECT:
diamond
schmitt

GENERAL CONTRACTOR:

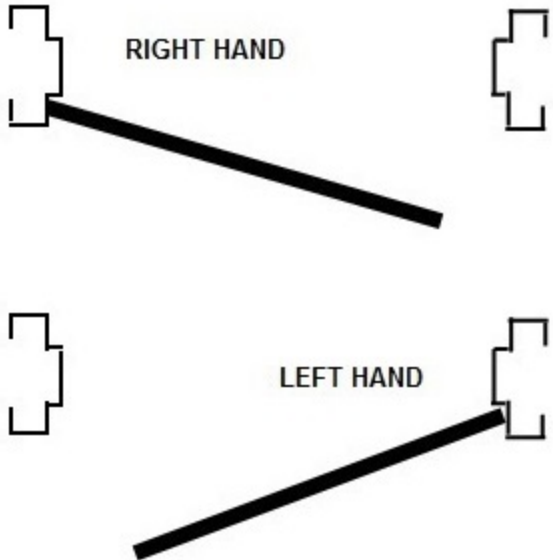
PROJECT COORDINATOR
EGRESS
SYSTEMS OF CANADA LTD

DATE: October 2 2025

SUBMISSION
2

DRAWN BY
DJL

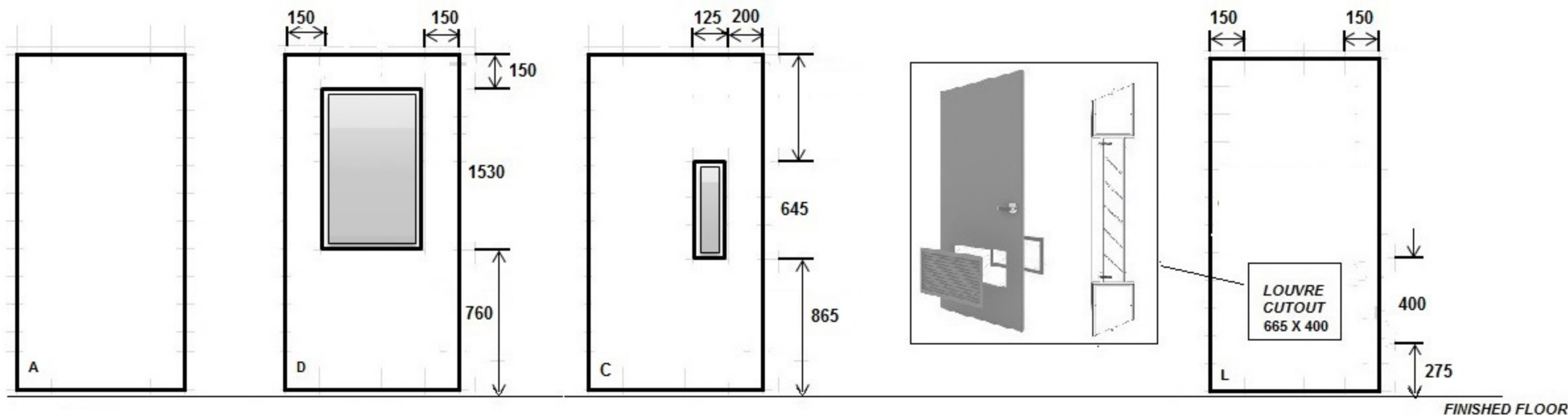
ES # ES25-015-002
S.O. # 45012



HOLLOW METAL FRAME AND DOOR SCHEDULE																											Job No: 45012						
	LOCATION				DOOR INFO								FRAME INFO														Revision Date: 17 Oct 2025 9:46						
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS				PEMKO		ARCH				MD	PEMKO	THERMAL BREAK		WALL		REMARKS						
											SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R	SWING	TYPE	FINISH	Ga	ELEV	SECTIONS	RAIN DRIP	WD	HT		THCK	ANCH				
1	D1A-STA	SC	1	RFI 007	J	FG	HM	45	HT	19		L	1065	x	2440	FG	W/C	16			L	SC08	W/C	16	S1		15.E.G			1065 X 2440		T/A	SPRINKLER PROTECTED MATERIAL TO BE ADVISED.
2	D1A-STB	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A					965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
3	D1A-STD	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	R	965	x	2440	VL	W/C	16			R	F2	W/C	16	A					965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
4	D1B-STD	FR	1	13MM GLASS	C	VL	HM	45	HT	19		L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A		17			965 X 2440		S/F	
5	D101C	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
6	D107	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A					965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
7	D108	FR	1		C		WD	45	INT	19		L	965	x	2440						L	F2	W/C	16	A		17			965 X 2440		S/F	
8	D109	FR	1		A		WD	45	INT	19		L	965	x	2440						L	F2	W/C	16	A		6			965 X 2440		C/S	
9	D116E	FR	1	13MM GLASS	C	VL	HM	45	HT	19		R	965	x	2440	VL	W/C	16			R	F2	W/C	16	A		10			965 X 2440		T/A	
10	D118C	FR	1	13MM GLASS	C	VL	HM	45	HT	19		L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
11	D119A	FR	1		C		WD	45	INT	19		R	965	x	2440						R	F2	W/C	16	A		13			965 X 2440		C/S	
12	D119B	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		13			965 X 2440		C/S	
13	D121	FR	1		A	F	HM	45	HT	19	0HR	L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
14	D122	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
15	D123	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
16	D125A	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
17	D125B	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
18	D126A	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
19	D126B	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
20	D127A	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
21	D130D	FR	1	13MM GLASS	D	G	HM	45	HT	19		L	965	x	2440	G	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
22	D131B	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
24	D2A-STB	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A					965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
25	D2A-STD	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A					965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
26	D2A-STE	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		13			965 X 2440		C/S	
27	D204	FR	1		A		WD	45	HT	19		L	965	x	2440						L	F2	W/C	16	A		17			965 X 2440		S/F	
28	D205	FR	1		A		WD	45	HT	19		L	965	x	2440						L	F2	W/C	16	A		17			965 X 2440		S/F	
29	D206	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		12			965 X 2440		C/S	
30	D207A	FR	1		L	F1	HM	45	HT	19		R	965	x	2440	F1	W/C	16			R	F2	W/C	16	A		12			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR
31	D207B	FR	1		L	F1	HM	45	HT	19		L	965	x	2440	F1	W/C	16			L	F2	W/C	16	A		13			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR
32	D208	FR	1		A		WD	45	HT	19		L	965	x	2440						L	F2	W/C	16	A		12			965 X 2440		C/S	
33	D209	FR	1		A		WD	45	INT	19		R	965	x	2440						R	F2	W/C	16	A		12			965 X 2440		C/S	
34	D213B	FR	1	212 TO 213	A		WD	45	HT	19		R	1220	x	2440						R	F2	W/C	16	A2		13F			1220 X 2440		C/S	OPEN CASE SLIDER FRAME
35	D215	FR	1		A		WD	45	HT	19		R	965	x	2440						R	F2	W/C	16	A		13			965 X 2440		C/S	
36	D215B	FR	1	WASH TO 215	A		WD	45	INT	19		R	915	x	2440						R	F2	W/C	16	A2		12F			915 X 2440		C/S	OPEN CASE SLIDER FRAME
37	D216	FR	1		A		WD	45	INT	19		R	965	x	2440						R	F2	W/C	16	A		13			965 X 2440		C/S	

HOLLOW METAL FRAME AND DOOR SCHEDULE																												Job No: 45012				
	LOCATION				DOOR INFO										FRAME INFO													Revision Date: 17 Oct 2025 9:46				
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	UND			CUT	LABEL	DOOR LEAF DETAILS				PEMKO		ARCH SWING	TYPE	FINISH	Ga	MD ELEV	PEMKO SECTIONS	THERMAL BREAK		WALL		REMARKS				
							MAT	THK				SW	WIDTH	HEIGHT	TYPE	FINISH	Ga							T/R	RAIN DRIP	WD	HT		THCK	ANCH		
38	D217	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		11			965 X 2440		C/S	
39	D219	FR	1		L	F1	HM	45	HT	19	0HR	L	965	x	2440	F1	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR
40	D220	FR	1	RFI 008 PENDING ELEV	C		WD	45	INT	19		L	965	x	2440					L	F2	W/C	16	A		13			965 X 2440		*	*C/S HINGE SIDE STRAP ANCHOR STRIKE SIDE
41	D221	FR	1		D		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S	
42	D222	FR	1		A		WD	45	INT	19		L	965	x	2440					L	F2	W/C	16	A		13			965 X 2440		C/S	
43	D223	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		13			1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16														
44	D224	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S	
45	D225	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		6			1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16														
46	D227	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	
47	D229A	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		13			1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16														
48	D229C	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	
49	D230	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S	
50	D231	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S	
51	D232	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	
52	D233	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S	
53	D235	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		12			965 X 2440		C/S	
54	D236	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S	
55	D237	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		12			965 X 2440		C/S	
56	D238	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S	
57	D239	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		12			965 X 2440		C/S	
58	D3A-STD	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	INT	19	3/4	R	965	x	2135	VL	W/C	18		R	F2	W/C	16	A					965 X 2135		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
59	D3B-STD	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		13			965 X 2135		T/A	
60	D302C	FR	1		A	F	HM	45	HT	19	3/4	L	1067	x	2440	F	W/C	16		PR	F2	W/C	16	B		13			2134 X 2440		T/A	
												R	1067	x	2440	F	W/C	16														
61	D303	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		12			965 X 2135		C/S	
62	D304	FR	1		A	F	HM	45	INT	19		L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		15			965 X 2135		C/S	
63	D305	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		12			965 X 2135		C/S	
64	D306	FR	1		A	F	HM	45	INT	19	3/4	L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		12			965 X 2135		C/S	
65	D307	FR	1		A	F	HM	45	INT	19	3/4	L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		12			965 X 2135		C/S	
66	D1A-STE	FR	1		A	F	HM	45	EXT	19		L	965	x	2440	F	G90	16		L	F1	G90	16	A		T8	☒		965 X 2440		S/F	
67	D1C-STD	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	
68	D101B	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	
69	D102F	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	
70	D116A	FR	1	13MM GLASS	C	VL	HM	45	EXT	19		R	965	x	2440	VL	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	

HOLLOW METAL FRAME AND DOOR SCHEDULE																								Job No: 45012
	LOCATION				DOOR INFO							FRAME INFO												Revision Date: 17 Oct 2025 9:46
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS SW WIDTH HEIGHT TYPE FINISH Ga T/R	ARCH SWING	TYPE	FINISH	Ga	MD ELEV	PEMKO SECTIONS	THERMAL BREAK RAIN DRIP		WD	HT	WALL THCK	ANCH	REMARKS
108	SC10-8	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
109	SC10-14	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
110	SC10-7	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
111	SC10-6	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
112	SC10-5	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
113	SC10-4	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
114	SC10-3	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
115	SC10-2	SC	1						INT				SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
116	SC14	SC	1						INT				SC14	W/C	16	S4		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
117	SC15	SC	1						INT				SC15	W/C	16	S5		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED



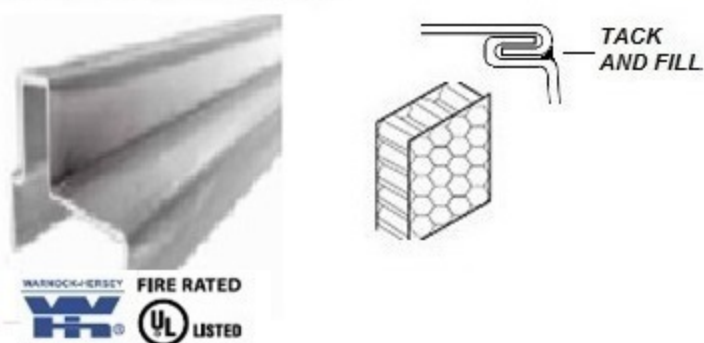
ELEV F
47 REQ'D

ELEV G
2 REQ'D
D130D, D130A
MAXIMUM GLASS THICKNESS
AT FIRE RATED DOORS 10MM

ELEV VL
12 REQ'D
MAXIMUM GLASS THICKNESS
AT FIRE RATED DOORS 10MM

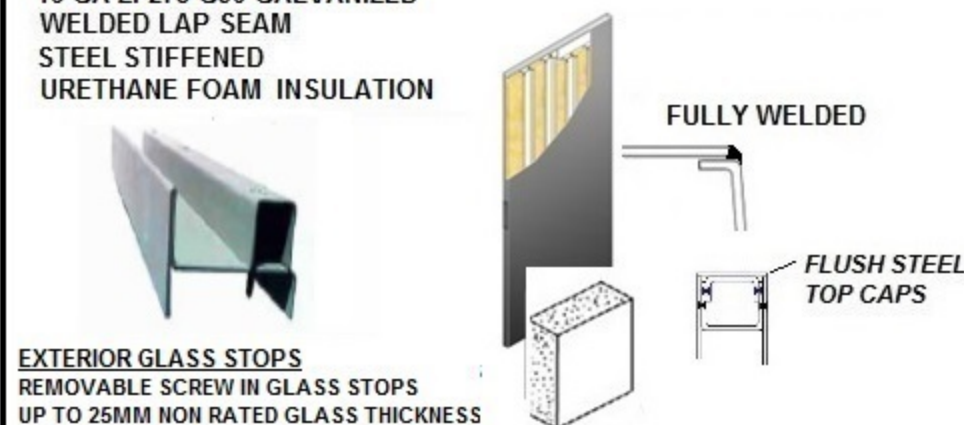
ELEV F1
3 REQ'D
D207B, D207A, D219

INTERIOR DOORS
18 GA A40 GALVANNEAL
WELDED AND FILLED INTERLOCKING SEAMS GROUND SMOOTH
HONEYCOMB BOARD INSULATION



INTERIOR SCREWED IN FIRE RATED GLASS STOPS
FOR RATED GLASS 7 TO 10MM THICK MAX.

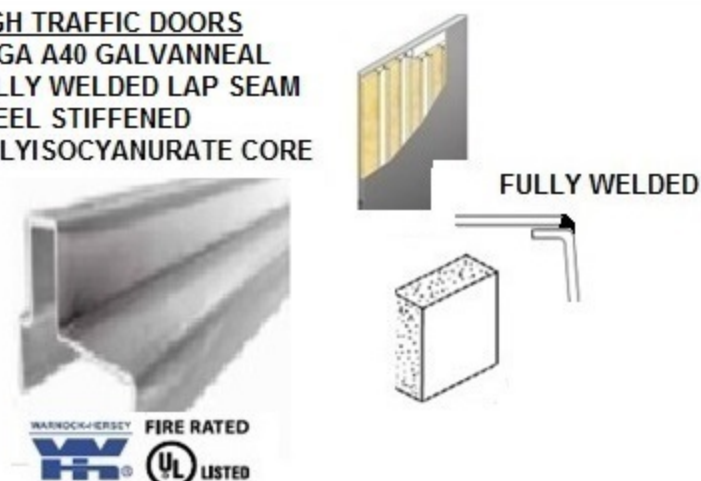
EXTERIOR DOORS
16 GA ZF275 G90 GALVANIZED
WELDED LAP SEAM
STEEL STIFFENED
URETHANE FOAM INSULATION



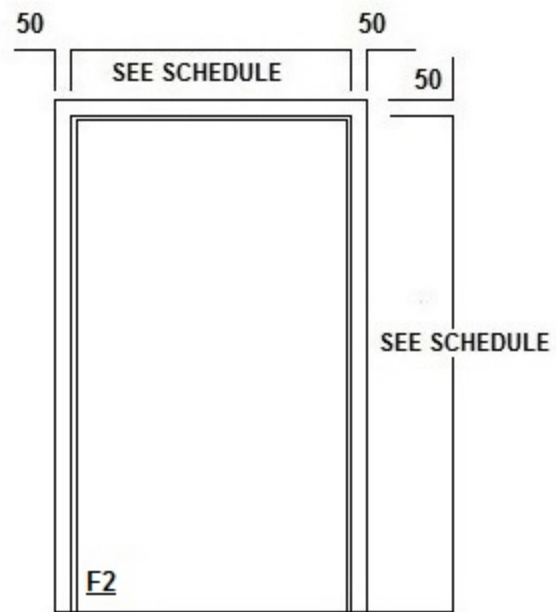
EXTERIOR GLASS STOPS
REMOVABLE SCREW IN GLASS STOPS
UP TO 25MM NON RATED GLASS THICKNESS

IT IS NOT RECOMMENDED TO PAINT EXTERIOR DOORS A DARK COLOR

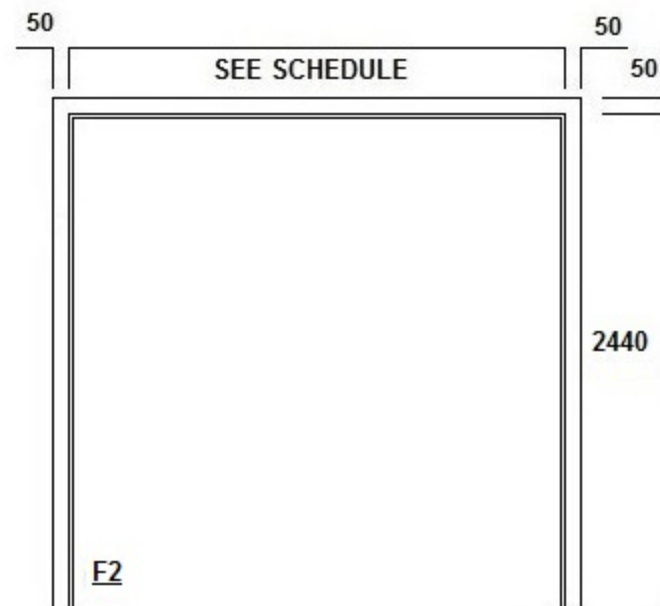
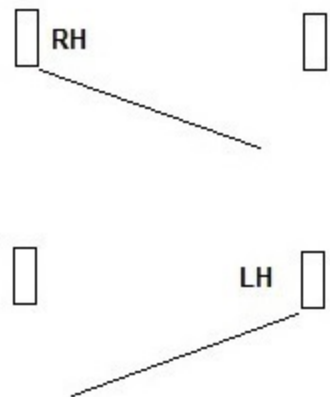
HIGH TRAFFIC DOORS
16 GA A40 GALVANNEAL
FULLY WELDED LAP SEAM
STEEL STIFFENED
POLYISOCYANURATE CORE



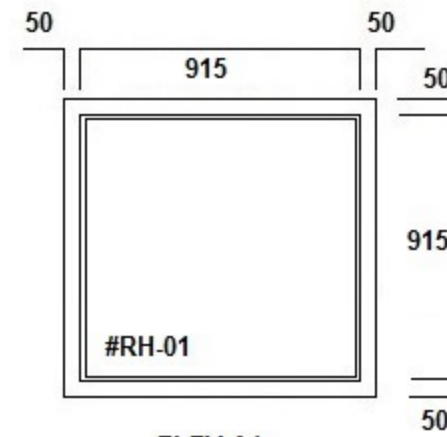
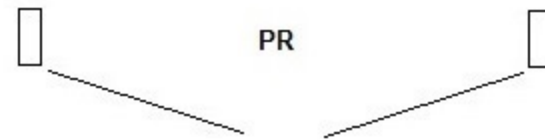
INTERIOR SCREWED IN FIRE RATED GLASS STOPS
FOR RATED GLASS 7 TO 10MM THICK MAX.



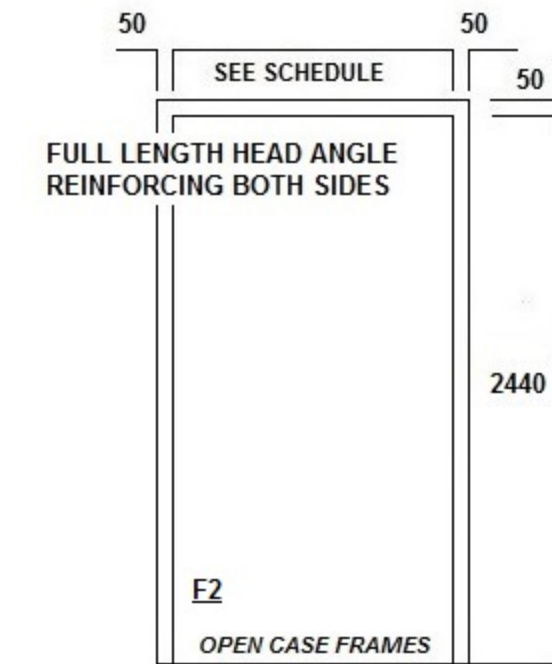
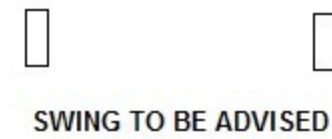
ELEV A
69 REQ'D



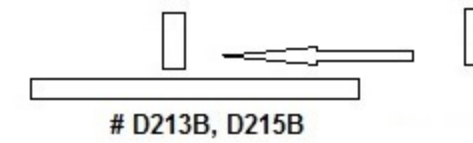
ELEV B
4 REQ'D

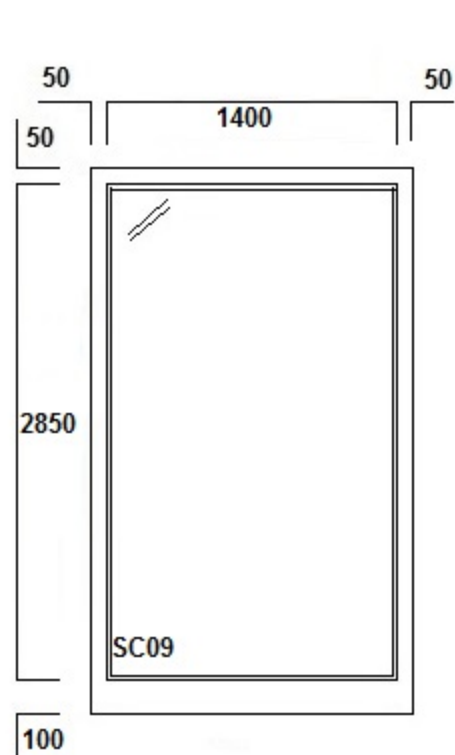


ELEV A1
1 REQ'D
4 SIDED ROOF HATCH
2 X S/F ON ALL 4 SIDES



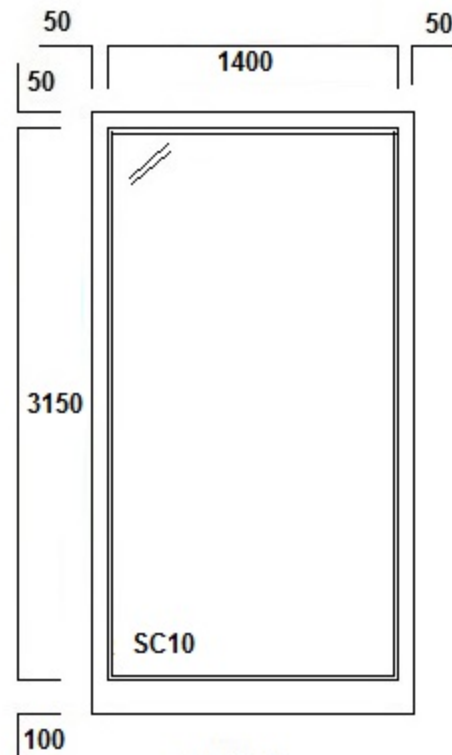
ELEV A2
2 REQ'D





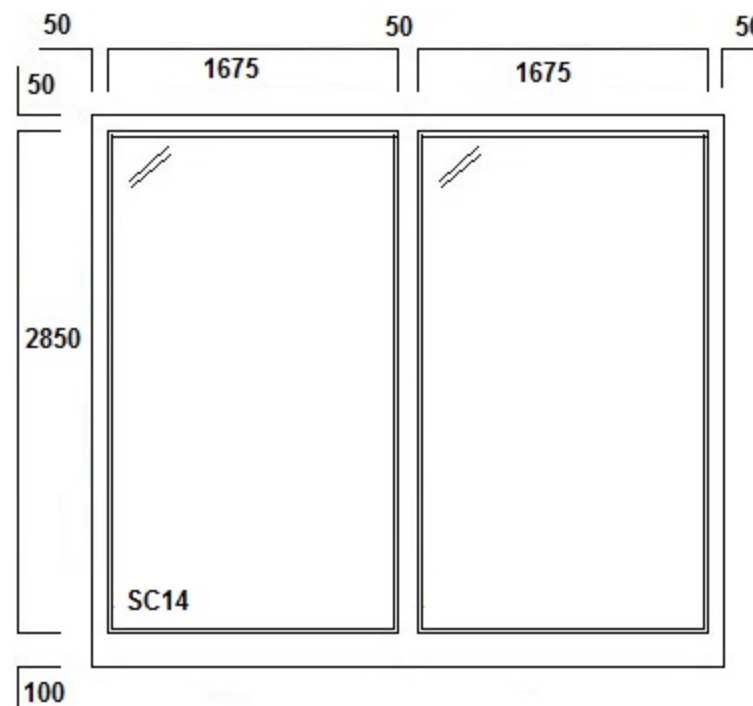
ELEV S2
21 REQ'D

SPRINKLER PROTECTED



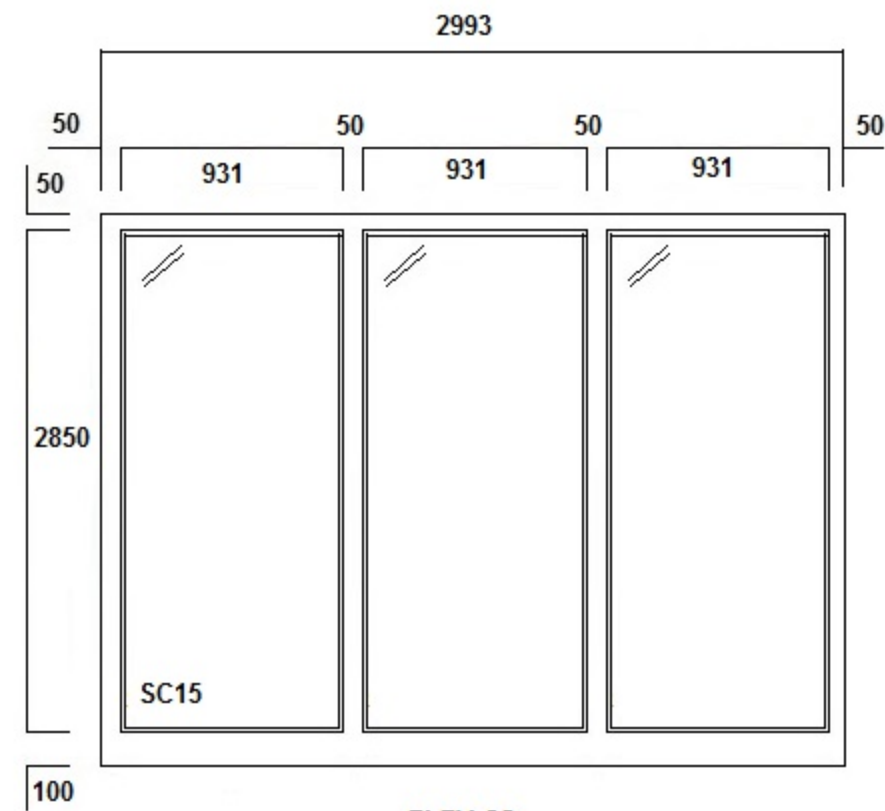
ELEV S3
14 REQ'D

SPRINKLER PROTECTED



ELEV S4
1 REQ'D

SPRINKLER PROTECTED

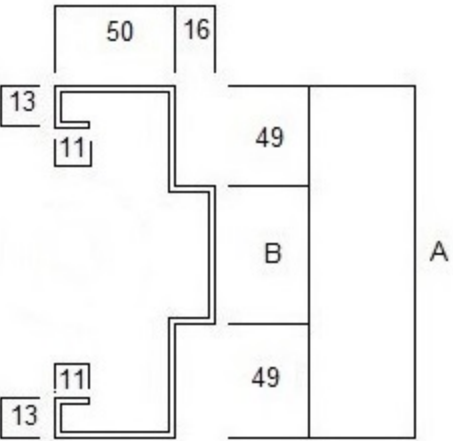


ELEV S5
1 REQ'D

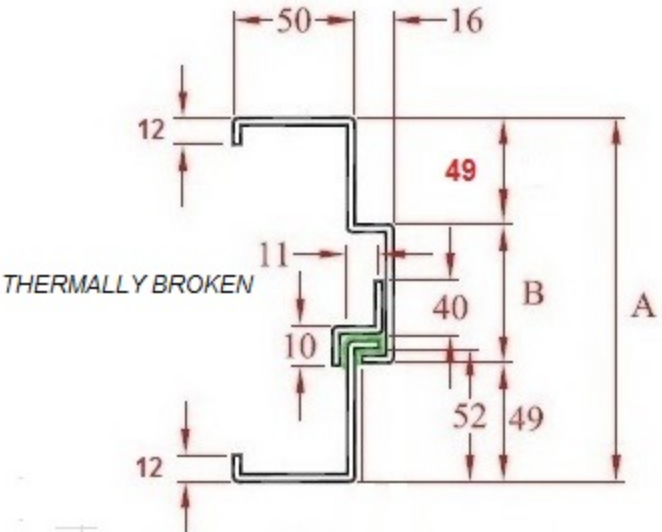
SPRINKLER PROTECTED

PLEASE CONFIRM ALL DIMENSIONS

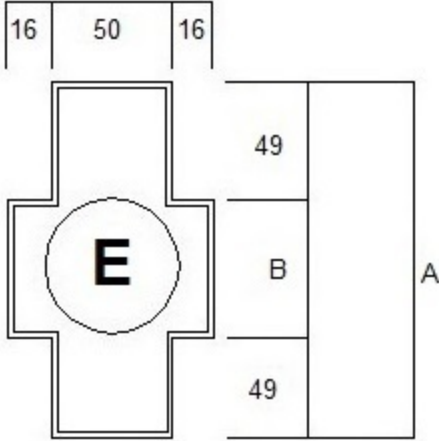
HOLLOW METAL SECTIONS



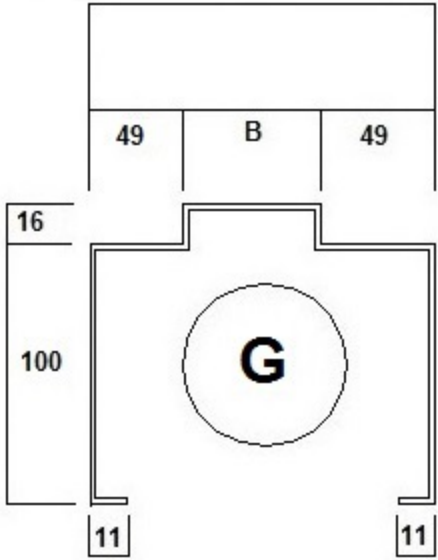
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE				
SECTION	A	B	GIRTH FRONT	BRAKE LINE
12	124	26		
13	140	42		
14	145	47		
15	150	52		
6	156	58		
8	190	92		
10	240	142		
15	265	167		
16	330	232	MAX JAMB DEPTH THAT CAN BE FIRE RATED.	
17	350	252		



"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE				
SECTION	A	B	GIRTH FRONT	BRAKE LINE
14T	145	47		
7T	184	86		
8T	190	92		
18T	250	152		

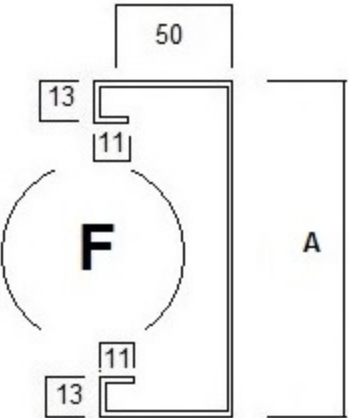


"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE				
SECTION	A	B	GIRTH FRONT	BRAKE LINE
15E	150	52		
6E	156	58		
8E	190	92		



"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE				
SECTION	A	B	GIRTH FRONT	BRAKE LINE
15G	150	52		
6G	156	58		
8G	190	92		

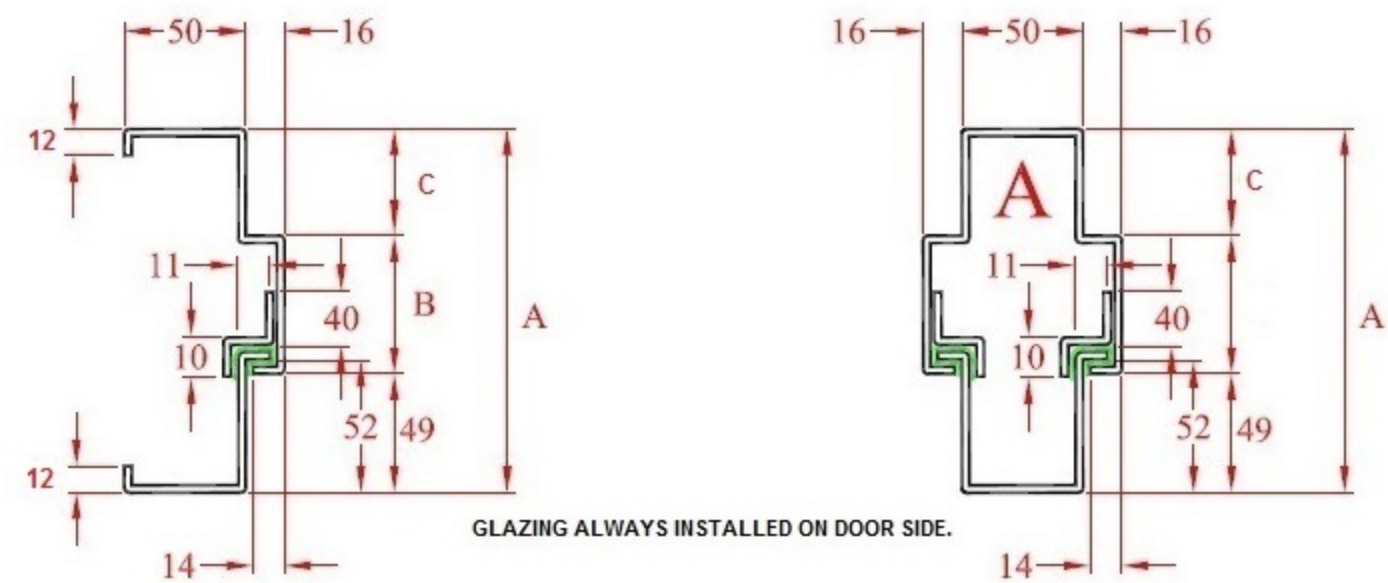
PLEASE CONFIRM ALL BASES TO BE 100mm



OPEN CASE FRAMES # A1-540A , A1-516A

"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE			
SECTION	A	GIRTH FRONT	BRAKE LINE
12F	124		
13F	140		

METAL DOOR LTD. TB SERIES
ALL THERMALLY BROKEN HOLLOW METAL PROFILES
THIS PAGE IS FOR REFERENCE ONLY
NOT FOR REVIEW.

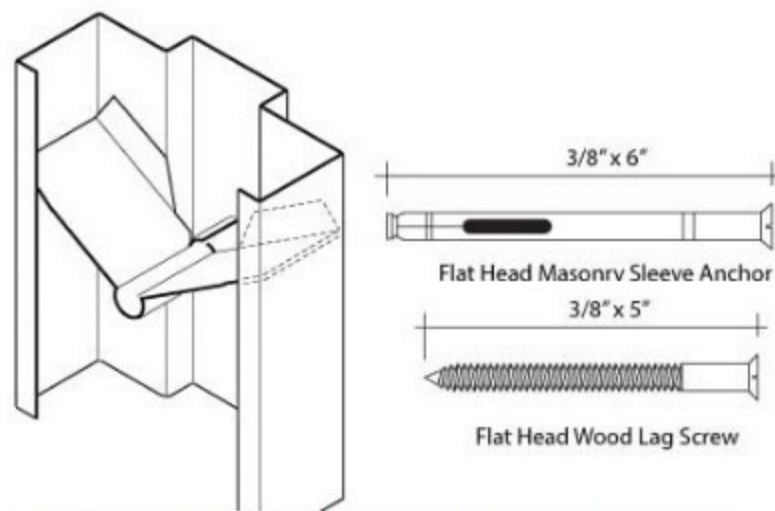
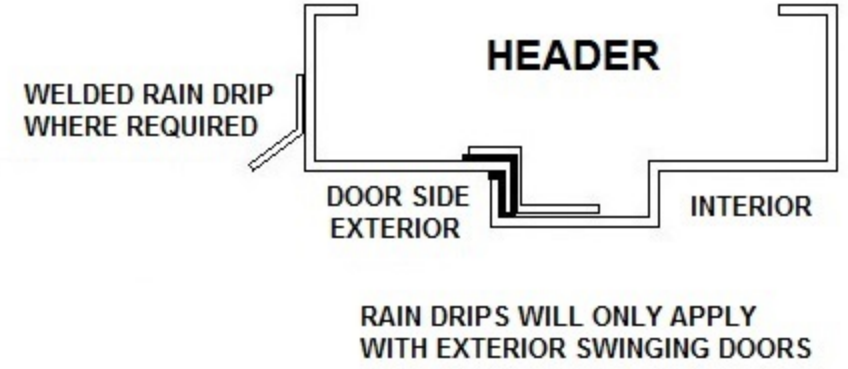
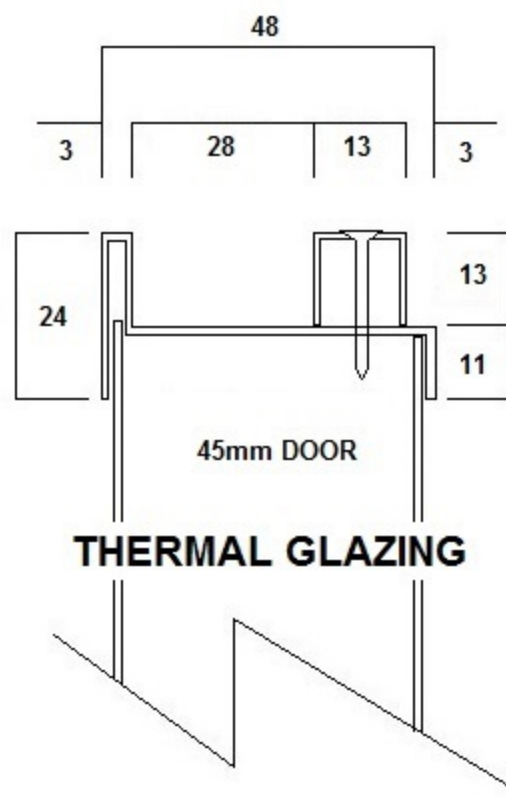


GLAZING ALWAYS INSTALLED ON DOOR SIDE.

SEE SECTION PAGE FOR ALL A, B, C DIMENSIONS

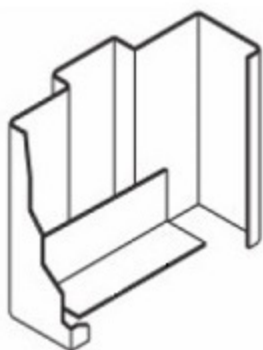
PLEASE SEE THE SECTION DETAILS FOR ALL
DIMENSIONS AND PROFILES REQUIRED

FOR SHIPPING PURPOSES
THERMALLY BROKEN SCREENS, WINDOWS OR TRANSOM FRAMES ARE GLAZED
WITH ADJUSTABLE GLASS STOPS FOR GLASS UP TO 25mm ALL SCREWS SHIP
COMPLETE WITH THE UNIT.
GLASS STOPS ARE ALL 18GA 16/19mm IS THE METAL DOOR LTD. STANDARD

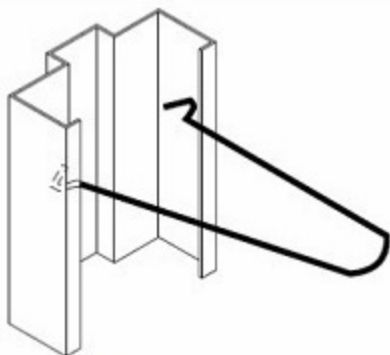


RECOMMENDED ANCHORAGE AT THERMALLY BROKEN
SCREW FIXED - EMA (S/F)
WITH GUIDE AND FRAME SUPPORT WELDED IN
3/8 FASTENERS SUPPLIED BY OTHERS

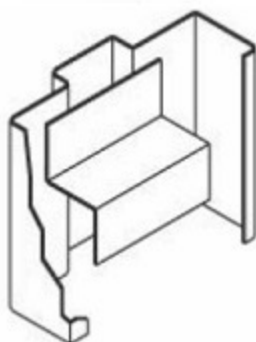
RECOMMENDED ANCHORAGE



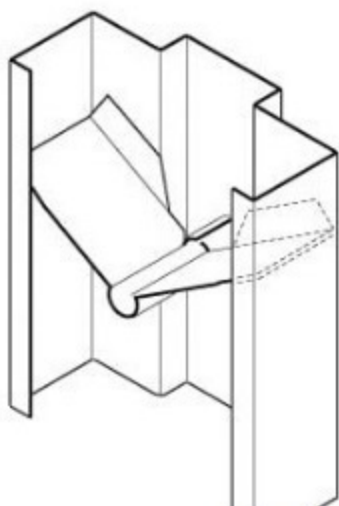
WELDED FLOOR ANCHOR (T/A)



MASONRY WIRE ANCHORS SHIPS LOOSE
WITH WELDED (T/A)



DRYWALL CHANNEL STUD ANCHOR (C/S)
WITH WELDED (T/A)



SCREW FIXED - EMA (S/F)
WITH GUIDE AND FRAME SUPPORT WELDED IN
3/8 FASTENERS SUPPLIED BY OTHERS



All Anchorage to be confirmed on shop drawings.
6 anchors per frame or screen up to 2286mm
8 anchors per frame or screen above 2286mm

Some anchors are shipped loose.

Extra or lost anchors can be ordered extra to contract.

STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

GUIDE SPECIFICATION FOR INSTALLATION AND STORAGE OF HOLLOW METAL DOORS AND FRAMES



Canadian Steel Door
Manufacturers Association

L'Association Canadienne
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STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

PART I - INTRODUCTION

This guide specification is intended to stress the necessary precautions and requirements for receipt, storage, handling and installation of hollow metal products, and the skills essential in the hanging of doors.

The proper performance of most products depends not only on how they are manufactured, but how they are installed. This is especially true of hollow metal doors and frames. The installation of doors and frames is an operation demanding care and skill, if the doors are to operate properly. Care in manufacturing does not, in itself, guarantee satisfactory performance. Even the best designed and most carefully constructed frames and doors, if incorrectly installed, will not function properly.

Hollow metal work is fabricated in accordance with the shop drawings, approved by the architect or engineer. Preparation for hardware or other items supplied by others is provided in accordance with the information furnished to the hollow metal manufacturer.

The hollow metal manufacturer is a material supplier, not a subcontractor. The manufacturer does not include the installation of their product in the building, but only shipment in good condition from the factory.

Should the General Contractor discover any error in the hollow metal delivered to the job site, it is imperative that the hollow metal manufacturer be notified in writing and allowed sufficient time before initiating any corrective measure in the field, so that the manufacturer can participate in solving the problem. Failure to do so could result in the cancellation of the warranty and/or fire label and non-acceptance of any cost associated with repair.

Most member companies of the Hollow Metal Manufacturer's Association Division of CSDMA have their own field representatives who are qualified not only to do expert repair work but to determine whether the fault lies with the manufacturer or with some other party.

It is essential that material is properly stored prior to installation and skills are exercised in the setting of frames and hanging of doors.

PART 2 - RECEIVING AND STORAGE OF MATERIAL

A. RECEIVING

Upon delivery, the contractor responsible for receiving hollow metal products shall thoroughly inspect for damage. Cardboard and other wrappings shall be removed for inspection and to promote air circulation. Any scratches or disfigurements caused in shipping or handling shall be promptly cleaned and touched up with a direct to metal rust inhibitive primer.

Should damaged material be found, the General Contractor has the option of refusing delivery or to accept the material as damaged. For coordination purposes, CSDMA suggests that delivery should not be refused, but rather accepted as damaged. Any damaged items should be noted on the freight bill. Claims will not be honored by the freight carrier, unless the damaged items are noted on the freight bill at the time of delivery. The contractor shall notify the hollow metal manufacturer in writing immediately of any item signed for as damaged.

B. ON SITE STORAGE

Proper storage of hollow metal work at the construction site will help prevent damage to the primer coat of paint. Prime coated steel must be protected when exposed to the elements such as high humidity, salt air, rain, snow, and/or damp wrappings etc.... Particular attention must, therefore, be given to steel products having a coat of factory applied primer. Primer is porous to properly receive and hold top coats. Water or moisture, in contact with primer coated steel will seep through to the steel. An electrolytic action then follows, resulting in corrosion and causing the paint film to lose adhesion. The presence of oxygen at the water-air interface behind the loosened paint film accelerates corrosive action and the prime coat further deteriorates.

Even when zinc coated steel is used to provide corrosion resistance, manufacturers of hollow metal door and frame products have found that one week of product exposure to water, due to improper storage, can be equivalent to at least a year of outdoor exposure to the elements.

NOTE: Paint manufacturers advise that the primer typically used by hollow metal manufacturers should receive a finish coat of paint within 30 days of delivery. It is the responsibility of the General Contractor to sand, touch up and clean prime painted surfaces prior to finish painting in accordance with the finish paint manufacturer's instructions.

The following procedures shall always be observed in storing hollow metal doors and frames at the job site:

1. Store all materials in a dry area, under cover. All products shall be stored where they will not be exposed to, or come in contact with the elements.
2. Do not use non-vented plastic or canvas. These materials create a humidity chamber, which promotes blistering and corrosion.
3. Store doors and frames in an upright position with heads uppermost, Figures 1 and 2.
4. Place no more than 5 doors or welded frames in a group. Small groups not only minimize the likelihood of damage due to excess handling, but also facilitate selection from the group for installation. In the case of multi-opening frames, no more than three units should be stored in a group, to avoid serious damage to the bottom most frame.
5. Place all material on planking or blocking at least 4 in. (100 mm) off the ground, 2 in. (50 mm) off a paved area or the floor slab.
6. Provide at least 1/4 in. (6.4 mm) space between all units to permit air circulation.

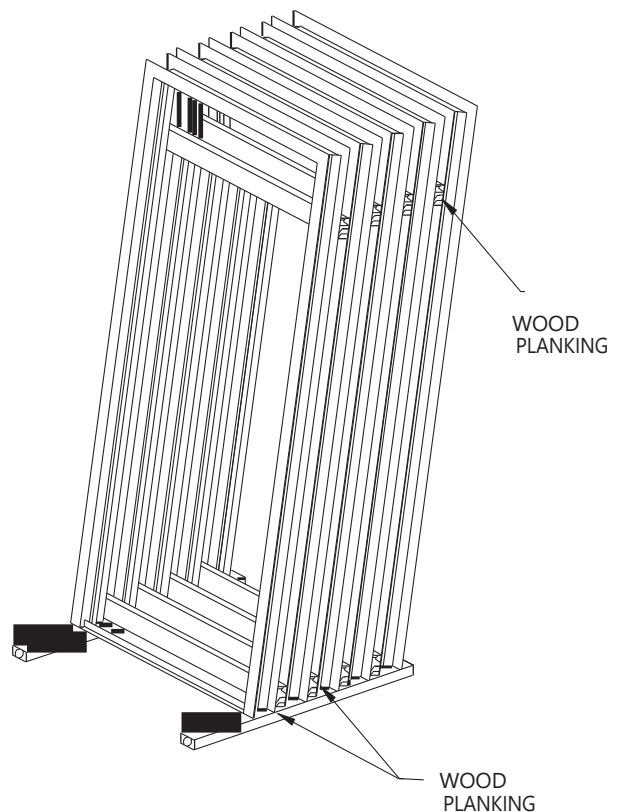


FIGURE 1
FRAME STORAGE

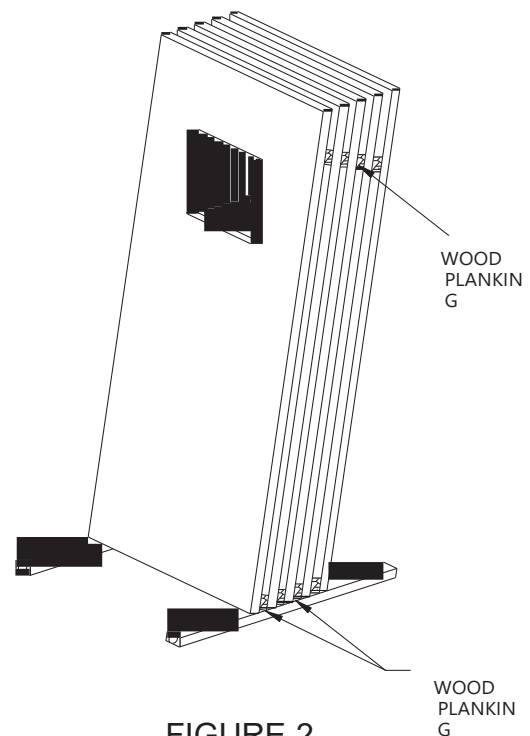


FIGURE 2
DOOR STORAGE

PART 3 - INSTALLATION OF FRAMES

A. GENERAL

Welded door frames are checked at the factory to ensure that they are square and that no jamb twists have occurred during fabrication. Temporary steel spreaders are then attached to the jamb base to minimize misalignment or other damage during handling and shipment. The frames are loaded on the carrier by personnel experienced in frame packing. The spreaders are for shipping and handling purposes only and must be removed before installing the frame.

At no time shall the spreaders be used during installation.

In spite of precautions, frames can and sometimes do arrive at the job site with minor alignment deviations. Minor deviations from true form and alignment can be corrected by the contractor responsible for installation.

When knock-down frames are used the manufacturer's recommended assembly procedures must be followed before installation of the frame can begin. A knock-down frame is easier to pack for shipment and requires less shipping space than a welded frame. Such frames, however, place greater responsibility on the installer, who must assemble them and make sure that, in spite of job conditions, they are square and in true alignment. Frames of this type normally use bendable tabs, screw or bolt fasteners in their assembly, and these may be lost in shipment or at the job site. In general, knockdown frames are not recommended for doors wider than 3 ft. 6 in. (1067 mm) or weighing more than 100 pounds (45 kg).

B. PRIOR TO INSTALLATION

The installer shall perform the following prior to installation: The

area of floor on which the frame is to be installed and the path of the door swing shall be checked for flatness and levelness. Permissible tolerance is +/- 1/16 in. (1.5mm). If the floor exceeds this, it is the general contractor's responsibility to correct the area that is out of tolerance before the frame is installed.

Frame shall be checked for correct size, swing, fire rating and opening number. Remove temporary steel spreaders. Spreader can typically be removed with cold chisel and hammer. With frame standing on concrete, position cold chisel at weld joint of spreader and jamb and strike chisel with hammer.

C. INSTALLATION TOLERANCES

During the setting of the frame, check and correct as necessary for opening width, opening height, squareness, alignment, twist and plumbness. Permissible frame product installation tolerances shall be maintained within the following limits: See Figure 3.

Opening width – Measured horizontally from rabbet to rabbet at top, middle and bottom of frame; +/- 1/32 in. (1mm).

Opening height – Measured vertically between the frame head rabbet and top of floor or bottom of frame minus jamb extension at each jamb and across the head; +/- 1/16 in. (1.5mm).

Squareness – Measured at rabbet on a line from jamb, perpendicular to frame head; not to exceed 1/16 in. (1.5mm).

Alignment – Measured at jambs on a horizontal line parallel to the plane of the face; not to exceed 1/16 in. (1.5mm).

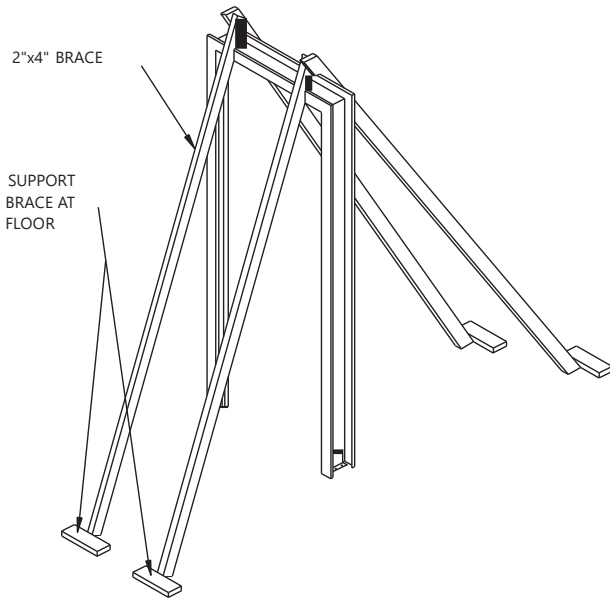
Twist – Measured at opposite face. Corners of jamb on parallel lines, perpendicular to the plane of the door rabbet.

Plumbness – Measured at jambs on a perpendicular line from the head to the floor; not to exceed 1/16 in. (1.5mm).

The tolerances shown provide a reasonable guideline for proper installation of hollow metal frame products. However, it should be noted that the cumulative effect of the installation tolerances at or near their maximum levels could result in sufficient misalignment to prevent the door from functioning properly. Installers should be careful not to create a tolerance buildup. Tolerance buildup occurs when several tolerances are at or near their maximums. Care should be taken to keep each of these tolerances as close to zero as possible.

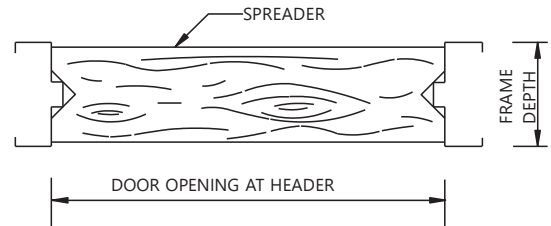
D. TYPICAL INSTALLATION PROCEDURES

Position frame in the correct location. Brace the frame as shown, Figure 4. Do not brace in the direction of intended wall.

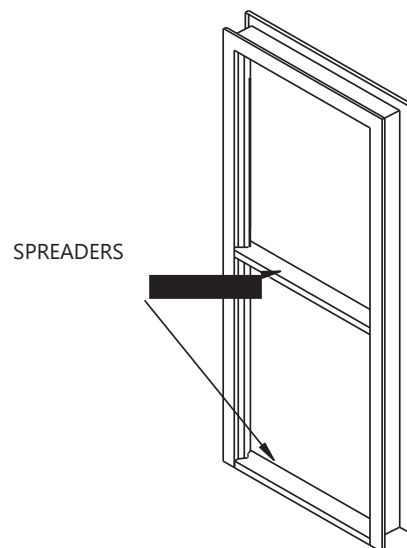


**FIGURE 4
FRAME BRACING**

With frame in position, install the temporary wood spreaders. The wood spreader, Figure 5, must be square and no less than 1 in. (25 mm) thick. Correct length is the door opening width between the jambs at the header. Cut clearance notches for frame stops. Spreader must be nearly as wide as frame depth for proper installation. Install a spreader at the bottom of the frame and a second wood spreader at the mid or strike point to maintain a proper door opening and to prevent bowing of the jambs, Figure 6. Clamp or wire spreaders to frame to hold spreaders in place until the frames are set permanently in the wall.



**FIGURE 5
WOOD SPREADER**



**FIGURE 6
SPREADER LOCATION**

At frames with jamb opening heights greater than 8 ft. (2438mm) or frame face dimensions less than 1-1/2 in. (38mm), install an additional wood spreader. Space wood spreaders at a maximum of 36 in. (914mm) intervals between header and bottom of frame, Figure 7.

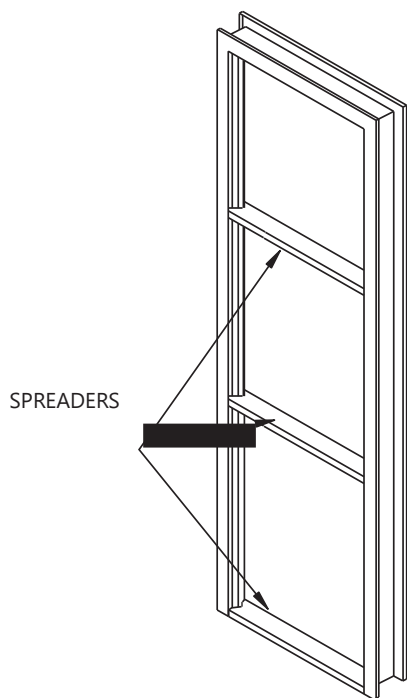


FIGURE 7
SPREADER LOCATION AT LARGE
JAMB OPENING HEIGHTS

At frames with sidelights where the sidelight sill intersects the door jamb near the strike, it is imperative that a wood spreader is located at this location, Figure 8.

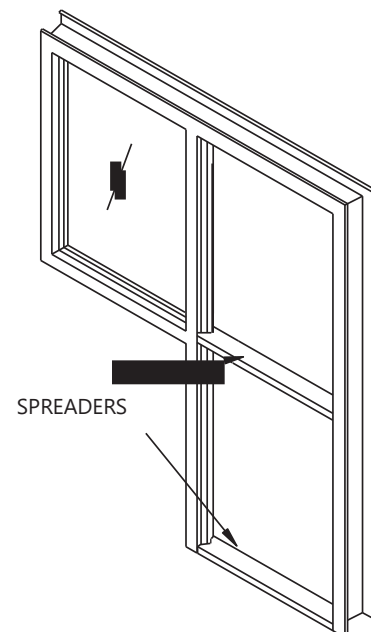
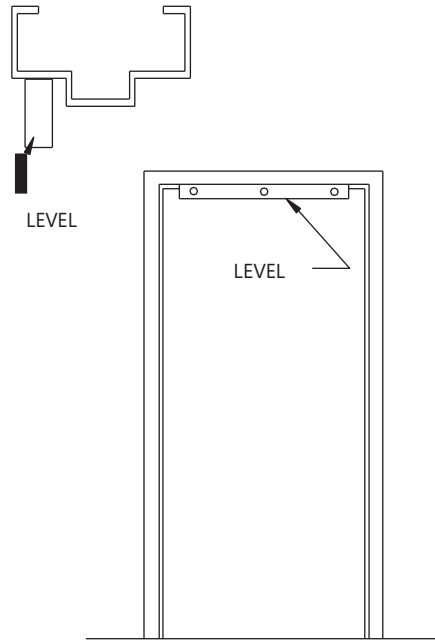


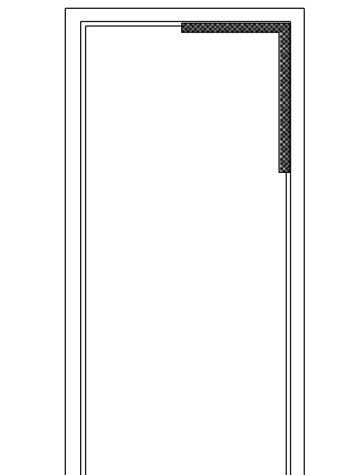
FIGURE 8
SPREADER LOCATION AT SIDELIGHTS

The installation contractor shall have a carpenter level and builder's square. Level the head by positioning the level to the head door rabbet, Figure 9. If necessary, adjust for high spots in floor by shimming under the jamb floor anchor, Figure 13a. Equalize them through an adjustable floor anchor, if specified, Figure 13b. Note, for labeled openings the maximum floor clearance is 3/4 in. (19mm).



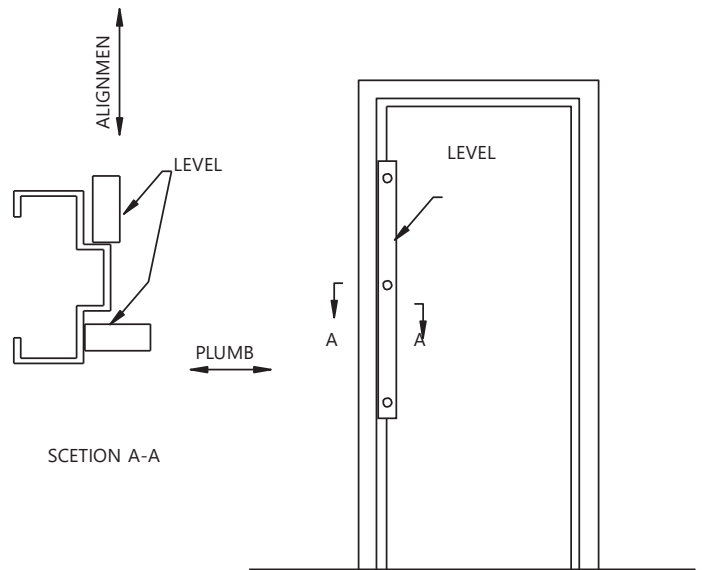
**FIGURE 9
LEVELING THE HEAD**

With builder's square, check frame for squareness. Position square against jamb and head at door rabbet, adjust as required, Figure 10.



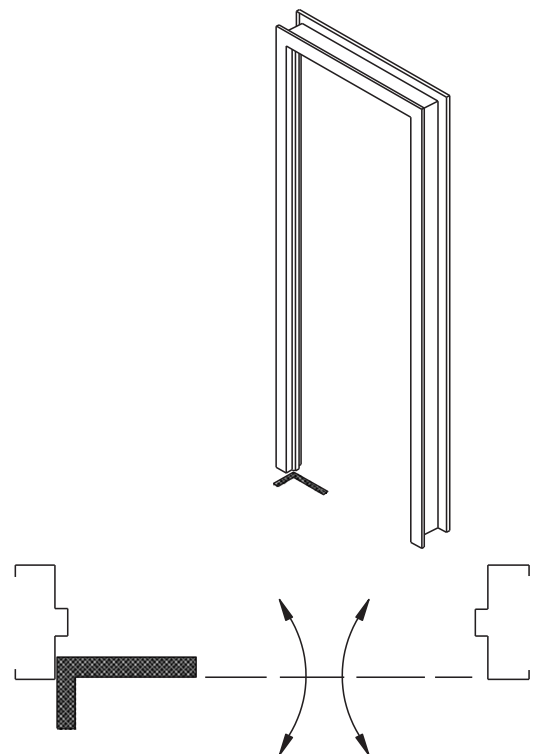
**FIGURE 10
SQUARENESS**

With carpenter level check the frame for plumbness and alignment: For plumbness, position level against both hinge and strike jambs in the rabbet. For alignment; Position level against both hinge and strike jambs on the stop, adjust as required, Figure 11.



**FIGURE 11
PLUMBING AND ALIGNMENT**

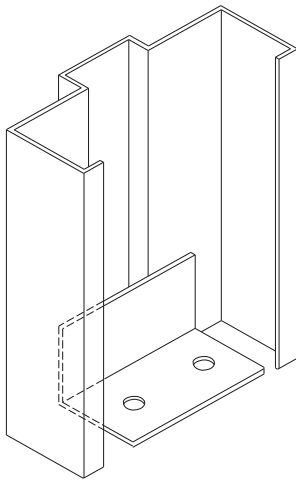
With builder's square, check jambs for twist. Position square against door rabbet and project line perpendicular to the plane of the door rabbet, adjust as required, Figure 12.



Once the installer has ensured that the frame is in the correct position, anchor the jamb to the floor.

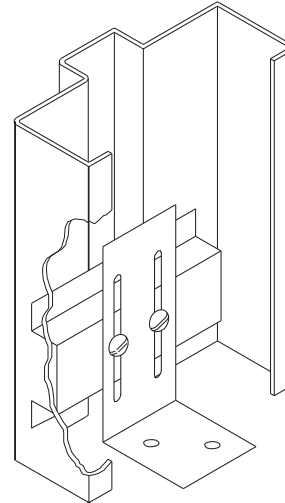
Floor Anchors: The fixed floor anchor is welded to the base of the jamb, typically secured to the floor by mechanical fasteners, providing solid anchorage for the base of each jamb, Figure 13a. Shimming must be used if the floor is not level. This type of anchor is not used in existing masonry, preframed stud walls or slip-on dry-wall frames, but may be provided.

At prepared openings (existing masonry or concrete) an additional expansion anchor reinforcement is provided. With pre-framed stud walls or completed drywall openings an additional base anchor is included. In both instances these anchors are located as close to the bottom of the jamb as is practical.



**FIGURE 13A
FIXED FLOOR ANCHOR**

The adjustable floor anchor is designed for use where there are significant irregularities from level, an intended slope in the floor, or where toppings such as terrazzo are used, Figure 13b. Like the standard anchor, it cannot be used in prepared openings or slip-on knock-down dry-wall frames.



**FIGURE 13B
ADJUSTABLE FLOOR ANCHOR**

E. WALL CONSTRUCTIONS AND APPROPRIATE ANCHORS

Proper frame anchoring is vitally important to the proper performance of the door opening. A variety of jamb wall anchors are available to suit the various types of wall construction. Anchors shown on the following pages are but a few of the variety of anchoring methods available.

ANCHOR SPACING

Anchor Spacing: CSDMA specifications require that the number of wall anchors used on each jamb, in all cases, be as follows:

- A. Openings in Masonry Walls. Frames with expansion bolt anchors; Anchors spaced a maximum of 6 in. (152 mm) from the top and bottom, with intermediate spacing at a maximum of 26 in. (660 mm) o/c. See Table 1.
- B. Openings in stud partitions with steel or wood stud anchors: Near hinges and directly opposite on strike jamb. See Table 1.
- C. Anchor quantity for slip-on drywall frames shall be as per manufacturer's standards and label requirements.
- D. Wall anchors locations and quantity are in addition to floor anchors, when required.

Table 1

Frame Anchor Spacing

Frame Height	Masonry Walls	Stud Partitions
> 0 in. < 60 in. (1524 mm)	2	3
> 60 in. < 90 in. (2286 mm)	3	4
> 90 in. < 96 in. (2438 mm)	4	5
> 96 in.	4 + 1 per 24 in. (610 mm)	5 + 1 per 24 in.



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STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

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STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

PART I - INTRODUCTION

This guide specification is intended to stress the necessary precautions and requirements for receipt, storage, handling and installation of hollow metal products, and the skills essential in the hanging of doors.

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It is essential that material is properly stored prior to installation and skills are exercised in the setting of frames and hanging of doors.

RELATED DOCUMENTS

- A. ANSI/A.250.11-2001 Recommended Erection Instructions for Steel Frames
- B. ANSI/NAAMM HMMA 801-05 Glossary of Terms for Hollow Metal Doors and Frames
- C. ANSI/NAAMM HMMA 841-07 Tolerances and Clearances for Commercial Hollow Metal Doors and Frames
- D. NAAMM HMMA 820-87 Hollow Metal Frames
- E. NFPA 80 Standard for Fire Doors and Other Opening Protectives 2007 Edition
- F. SDI-122-90 Installation & Troubleshooting for Standard Steel Doors and Frames

PART 2 - RECEIVING AND STORAGE OF MATERIAL

A. RECEIVING

Upon delivery, the contractor responsible for receiving hollow metal products shall thoroughly inspect for damage. Cardboard and other wrappings shall be removed for inspection and to promote air circulation. Any scratches or disfigurements caused in shipping or handling shall be promptly cleaned and touched up with a direct to metal rust inhibitive primer.

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B. ON SITE STORAGE

Proper storage of hollow metal work at the construction site will help prevent damage to the primer coat of paint. Prime coated steel must be protected when exposed to the elements such as high humid-

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The following procedures shall always be observed in storing hollow metal doors and frames at the job site:

1. Store all materials in a dry area, under cover. All products shall be stored where they will not be exposed to, or come in contact with the elements.
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3. Store doors and frames in an upright position with heads uppermost, Figures 1 and 2.
4. Place no more than 5 doors or welded frames in a group. Small groups not only minimize the likelihood of damage due to excess handling, but also facilitate selection from the group for installation. In the case of multi-opening frames, no more than three units should be stored in a group, to avoid serious damage to the bottom most frame.
5. Place all material on planking or blocking at least 4 in. (100 mm) off the ground, 2 in. (50 mm) off a paved area or the floor slab.
6. Provide at least 1/4 in. (6.4 mm) space between all units to permit air circulation.

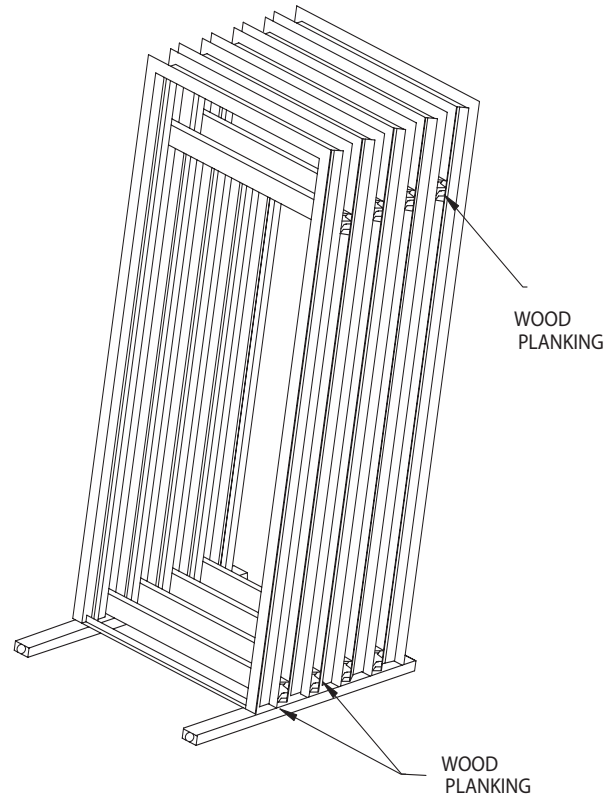


FIGURE 1
FRAME STORAGE

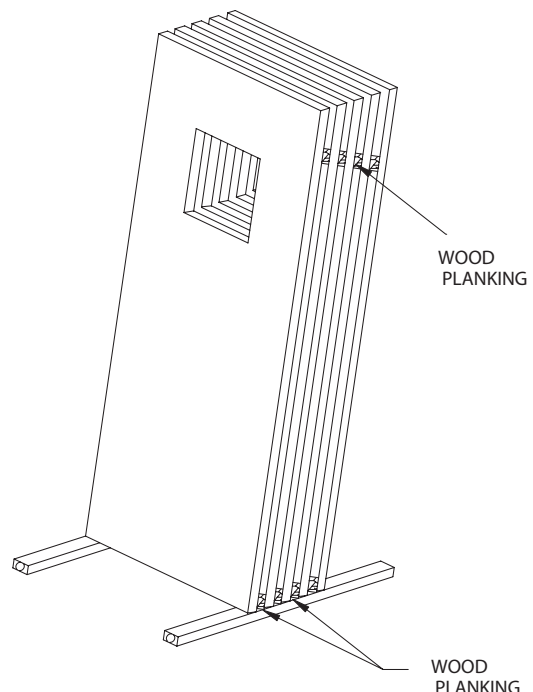


FIGURE 2
DOOR STORAGE

PART 3 - INSTALLATION OF FRAMES

A. GENERAL

Welded door frames are checked at the factory to ensure that they are square and that no jamb twists have occurred during fabrication. Temporary steel spreaders are then attached to the jamb base to minimize misalignment or other damage during handling and shipment. The frames are loaded on the carrier by personnel experienced in frame packing. The spreaders are for shipping and handling purposes only and must be removed before installing the frame.

At no time shall the spreaders be used during installation.

In spite of precautions, frames can and sometimes do arrive at the job site with minor alignment deviations. Minor deviations from true form and alignment can be corrected by the contractor responsible for installation.

When knock-down frames are used the manufacturer's recommended assembly procedures must be followed before installation of the frame can begin. A knock-down frame is easier to pack for shipment and requires less shipping space than a welded frame. Such frames, however, place greater responsibility on the installer, who must assemble them and make sure that, in spite of job conditions, they are square and in true alignment. Frames of this type normally use bendable tabs, screw or bolt fasteners in their assembly, and these may be lost in shipment or at the job site. In general, knock-down frames are not recommended for doors wider than 3 ft. 6 in. (1067 mm) or weighing more than 100 pounds (45 kg).

B. PRIOR TO INSTALLATION

The installer shall perform the following prior to installation:

The area of floor on which the frame is to be installed and the path of the door swing shall be checked for flatness and levelness. Permissible tolerance is $\pm 1/16$ in. (1.5mm). If the floor exceeds this, it is the general contractor's responsibility to correct the area that is out of tolerance before the frame is installed.

Frame shall be checked for correct size, swing, fire rating and opening number.

Remove temporary steel spreaders. Spreaders can typically be removed with cold chisel and hammer. With frame standing on concrete, position cold chisel at weld joint of spreader and jamb and strike chisel with hammer.

C. INSTALLATION TOLERANCES

During the setting of the frame, check and correct as necessary for opening width, opening height, squareness, alignment, twist and plumbness. Permissible frame product installation tolerances shall be maintained within the following limits: See Figure 3.

Opening width – Measured horizontally from rabbet to rabbet at top, middle and bottom of frame; $\pm 1/16$ in. (1.5mm).

Opening height – Measured vertically between the frame head rabbet and top of floor or bottom of frame minus jamb extension at each jamb and across the head; $\pm 3/64$ in. (1.2mm).

Squareness – Measured at rabbet on a line from jamb, perpendicular to frame head; not to exceed $1/16$ in. (1.5mm).

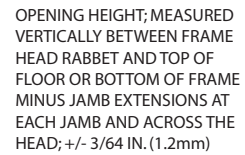
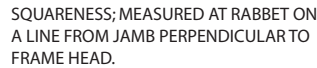
Alignment – Measured at jambs on a horizontal line parallel to the plane of the face; not to exceed $1/16$ in. (1.5mm).

Twist – Measured at opposite face. Corners of jamb on parallel lines, perpendicular to the plane of the door rabbet.

Plumbness – Measured at jambs on a perpendicular line from the head to the floor; not to exceed $1/16$ in. (1.5mm).

The tolerances shown provide a reasonable guideline for proper installation of hollow metal frame products. However, it should be noted that the cumulative effect of the installation tolerances at or near their maximum levels could result in sufficient misalignment to prevent the door from functioning properly. Installers should be careful not to create a tolerance buildup. Tolerance buildup occurs when several tolerances are at or near their maximums. Care should be taken to keep each of these tolerances as close to zero as possible.

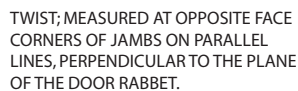
← MIDDLE AND BOTTOM OF FRAME; +/- 1/16
IN (1.5mm) →



ALIGNMENT; MEASURED AT JAMBS
ON A HORIZONTAL LINE PARALLEL TO
THE PLANE OF THE FACE.

1" / 16

1" / 16



FRAME INSTALLATION TOLERANCES

D. TYPICAL INSTALLATION PROCEDURES

Position frame in the correct location. Brace the frame as shown, Figure 4. Do not brace in the direction of intended wall.

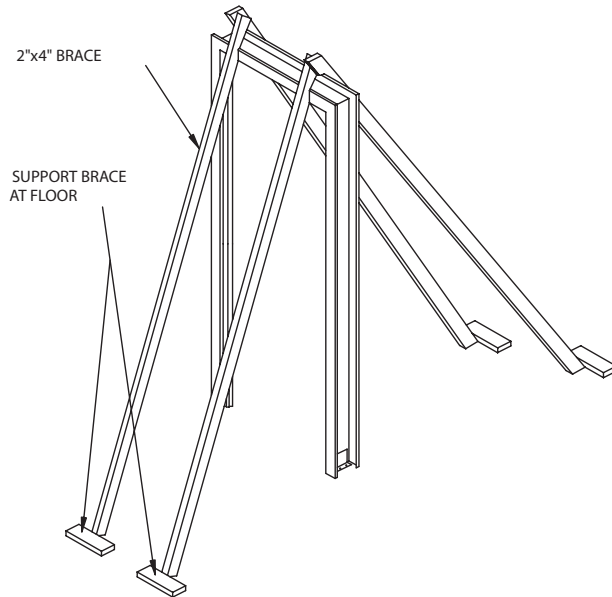


FIGURE 4
FRAME BRACING

With frame in position, install the temporary wood spreaders. The wood spreader, Figure 5, must be square and no less than 1 in. (25 mm) thick. Correct length is the door opening width between the jambs at the header. Cut clearance notches for frame stops. Spreader must be nearly as wide as frame depth for proper installation. Install a spreader at the bottom of the frame and a second wood spreader at the mid or strike point to maintain a proper door opening and to prevent bowing of the jambs, Figure 6. Clamp or wire spreaders to frame to hold spreaders in place until the frames are set permanently in the wall.

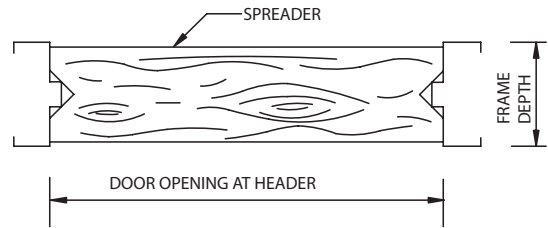


FIGURE 5
WOOD SPREADER

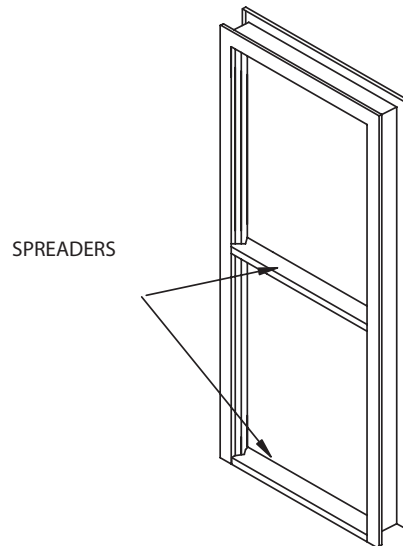


FIGURE 6
SPREADER LOCATION

At frames with jamb opening heights greater than 8 ft. (2438mm) or frame face dimensions less than 1-1/2 in. (38mm), install an additional wood spreader. Space wood spreaders at a maximum of 36 in. (914mm) intervals between header and bottom of frame, Figure 7.

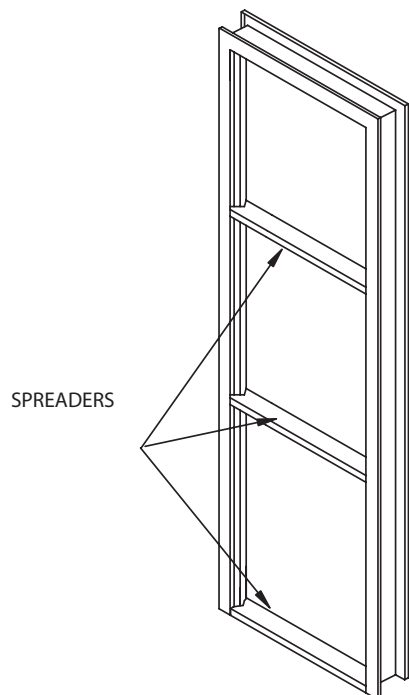


FIGURE 7
SPREADER LOCATION AT
LARGE JAMB OPENING HEIGHTS

At frames with sidelights where the sidelight sill intersects the door jamb near the strike, it is imperative that a wood spreader is located at this location, Figure 8.

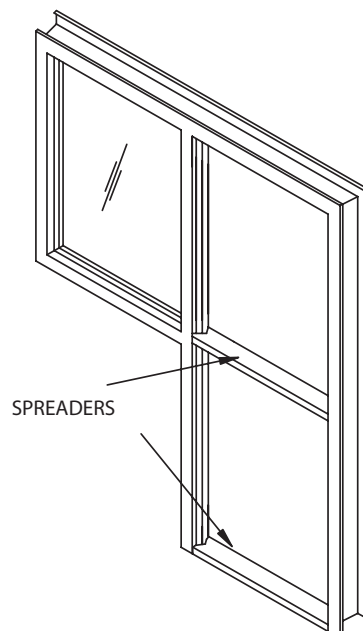
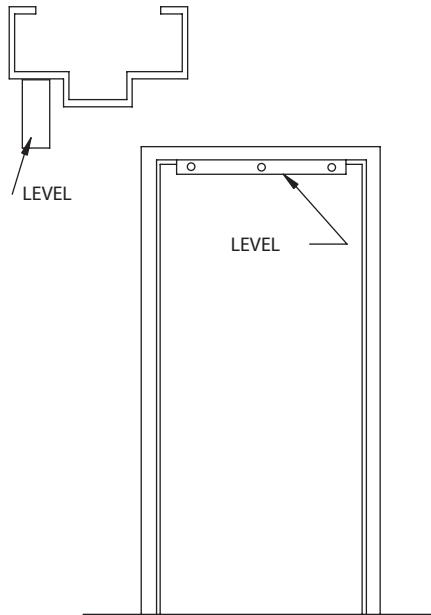


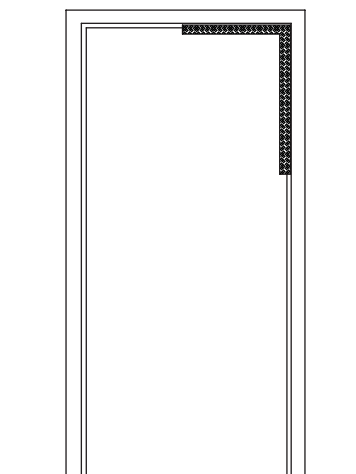
FIGURE 8
SPREADER LOCATION AT SIDELIGHTS

The installation contractor shall have a carpenter level and builder's square. Level the head by positioning the level to the head door rabbet, Figure 9. If necessary, adjust for high spots in floor by shimming under the jamb floor anchor, Figure 13a. Equalize them through an adjustable floor anchor, if specified, Figure 13b. Note, for labeled openings the maximum floor clearance is 3/4 in. (19mm).



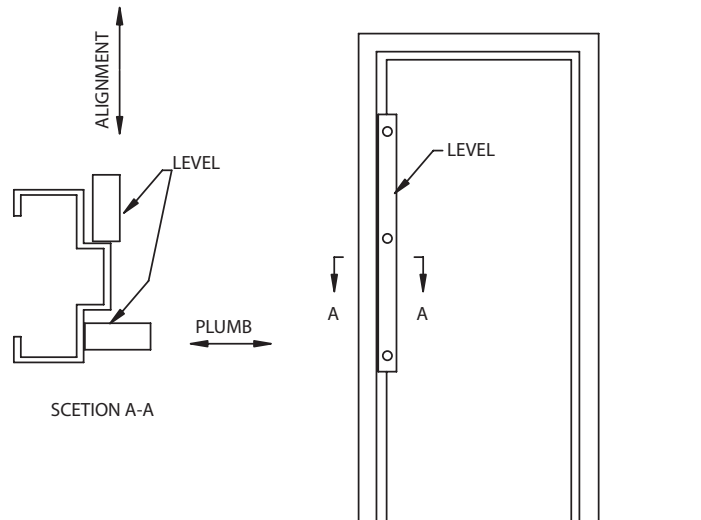
**FIGURE 9
LEVELING THE HEAD**

With builder's square, check frame for squareness. Position square against jamb and head at door rabbet, adjust as required, Figure 10.



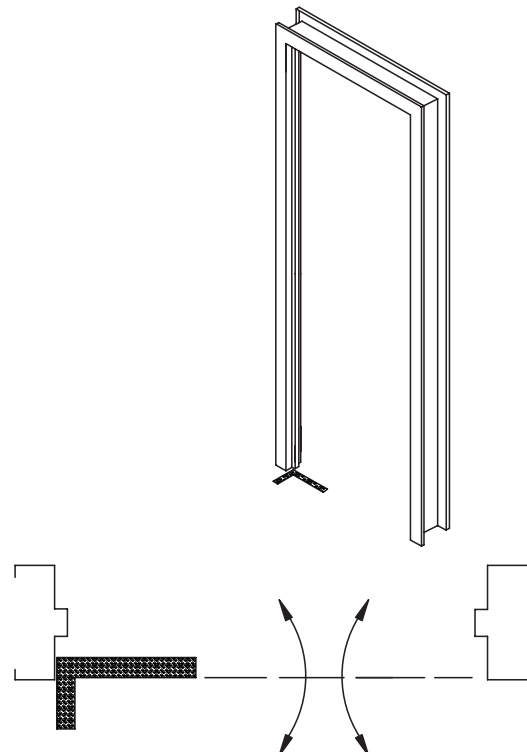
**FIGURE 10
SQUARENESS**

With carpenter level check the frame for plumbness and alignment: For plumbness, position level against both hinge and strike jambs in the rabbet. For alignment; Position level against both hinge and strike jambs on the stop, adjust as required, Figure 11.



**FIGURE 11
PLUMBING AND ALIGNMENT**

With builder's square, check jambs for twist. Position square against door rabbet and project line perpendicular to the plane of the door rabbet, adjust as required, Figure 12.



**FIGURE 12
FRAME TWIST CHECK**

Once the installer has ensured that the frame is in the correct position, anchor the jamb to the floor.

Floor Anchors: The fixed floor anchor is welded to the base of the jamb, typically secured to the floor by mechanical fasteners, providing solid anchorage for the base of each jamb, Figure 13a. Shimming must be used if the floor is not level. This type of anchor is not used in existing masonry, preframed stud walls or slip-on dry-wall frames, but may be provided.

At prepared openings (existing masonry or concrete) an additional expansion anchor reinforcement is provided. With pre-framed stud walls or completed drywall openings an additional base anchor is included. In both instances these anchors are located as close to the bottom of the jamb as is practical.

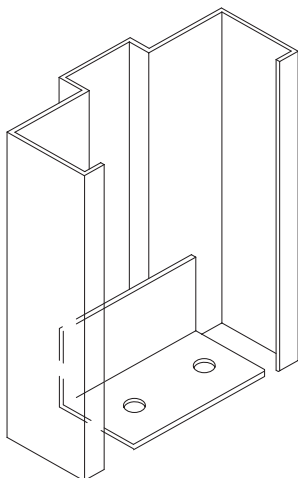


FIGURE 13A
FIXED FLOOR ANCHOR

The adjustable floor anchor is designed for use where there are significant irregularities from level, an intended slope in the floor, or where toppings such as terrazzo are used, Figure 13b. Like the standard anchor, it cannot be used in prepared openings or slip-on knock-down dry-wall frames.

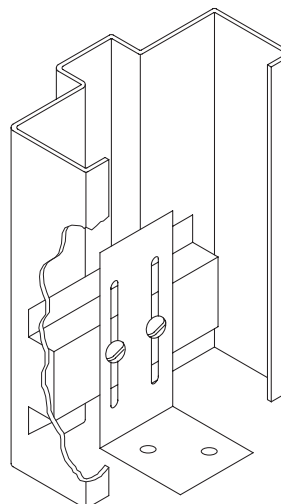


FIGURE 13B
ADJUSTABLE FLOOR ANCHOR

E. WALL CONSTRUCTIONS AND APPROPRIATE ANCHORS

Proper frame anchoring is vitally important to the proper performance of the door opening. A variety of jamb wall anchors are available to suit the various types of wall construction. Anchors shown on the following pages are but a few of the variety of anchoring methods available. For additional details, refer to HMMA 820 Hollow Metal Frames.

ANCHOR SPACING

Anchor Spacing: HMMA specifications require that the number of wall anchors used on each jamb, in all cases, be as follows:

- A. Openings in Masonry Walls. Frames with expansion bolt anchors; Anchors spaced a maximum of 6 in. (152 mm) from the top and bottom, with intermediate spacing at a maximum of 26 in. (660 mm) o/c. See Table 1.
- B. Openings in stud partitions with steel or wood stud anchors: Near hinges and directly opposite on strike jamb. See Table 1.
- C. Anchor quantity for slip-on drywall frames shall be as per manufacturer's standards and label requirements.
- D. Wall anchors locations and quantity are in addition to floor anchors, when required.

Table 1 Frame Anchor Spacing		
Frame Height	Masonry Walls	Stud Partitions
> 0 in. < 60 in. (1524 mm)	2	3
> 60 in. < 90 in. (2286 mm)	3	4
> 90 in. < 96 in. (2438 mm)	4	5
> 96 in.	4 + 1 per 24 in. (610 mm)	5 + 1 per 24 in.

1. MASONRY WALLS

Installation of a hollow metal frame in masonry, Figure 14a. Refer to HMMA 820 Hollow Metal Frames for additional anchorage methods.

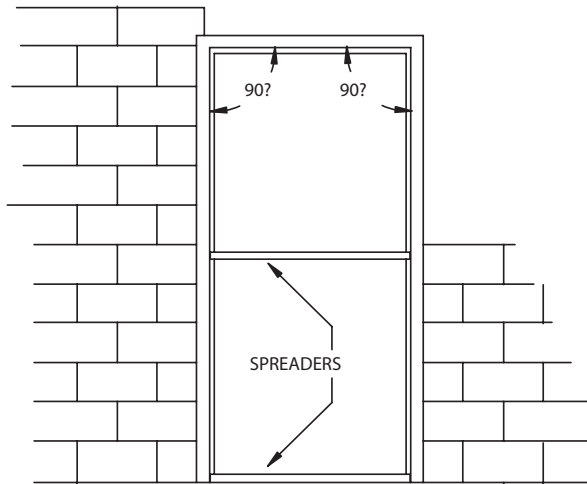


FIGURE 14A
MASONRY WALL

Adjustable strap-and-stirrup anchor, Figure 14b. A stirrup is welded to the back of the jamb, and a perforated or corrugated strap provides for embedding into masonry joints. The stirrup provides sufficient adjustment for masonry coursing as well as access for full grouting of jambs.

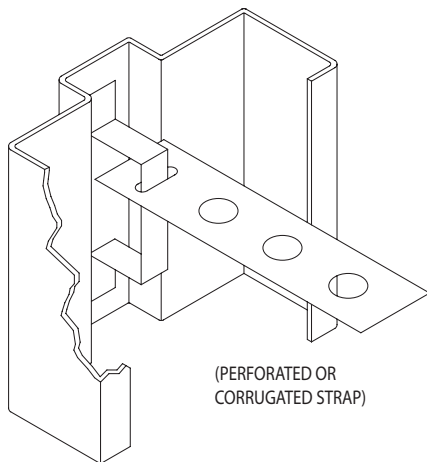


FIGURE 14B
STRAP & STIRRUP ANCHOR

T-strap anchor, Figure 14c and Wire Loop anchor, Figure 14d, function similar to the strap and stirrup anchor but are set loose inside frame returns as wall is laid up.

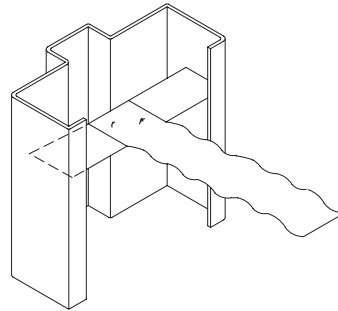


FIGURE 14C
T-STRAP ANCHOR

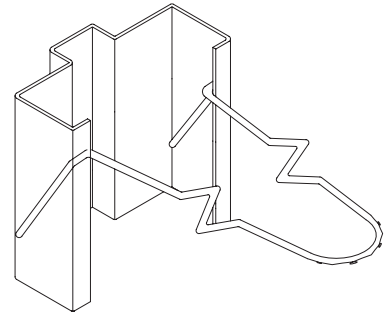


FIGURE 14D
WIRE LOOP ANCHOR

As wall is laid up, locate the anchors at the hinges in the hinge jamb and at a corresponding position in the strike jamb, Figure 14e.

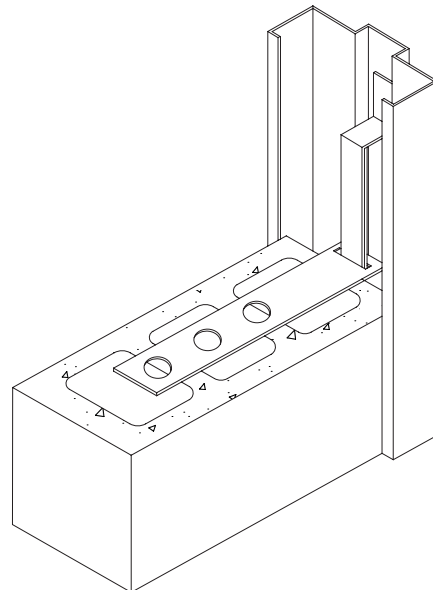


FIGURE 14E
MASONRY WALL WITH ANCHOR

Continually check squareness, plumbness, alignment, and twist in the frame as wall progresses.

2. STEEL STUD WALLS

Installation of a hollow metal frame in steel stud wall, Figure 15a. Refer to HMMA 820 Hollow Metal Frames for additional anchorage methods.

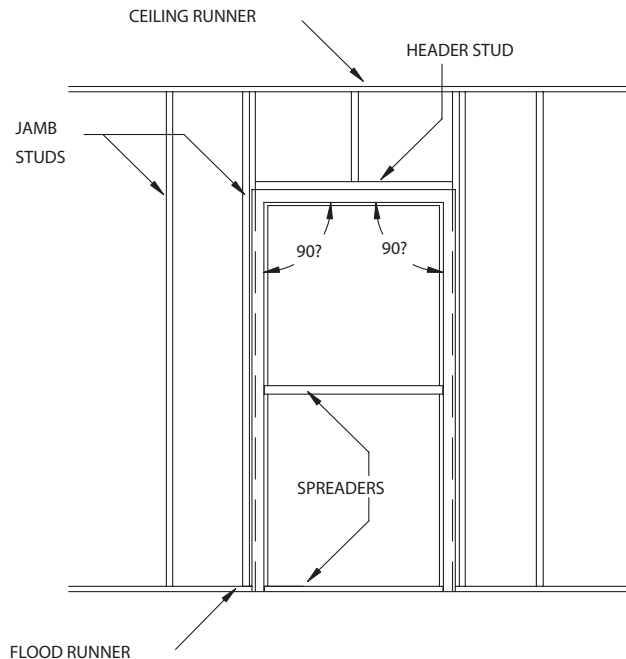


FIGURE 15A
STEEL STUD WALL

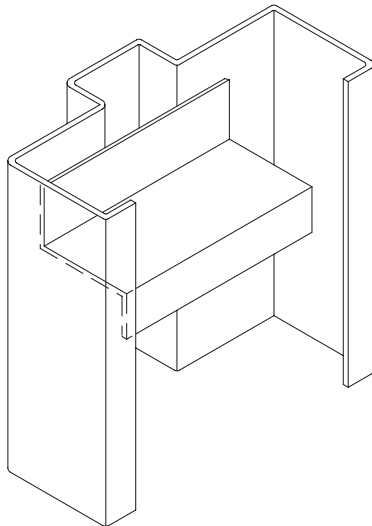


FIGURE 15B
ZEE SHAPED STEEL STUD ANCHOR

Zee shaped anchor, Figure 15b, is welded to both rabbets inside the jamb. Steel studs are mechanically fastened to the exposed flange through the throat of the stud.

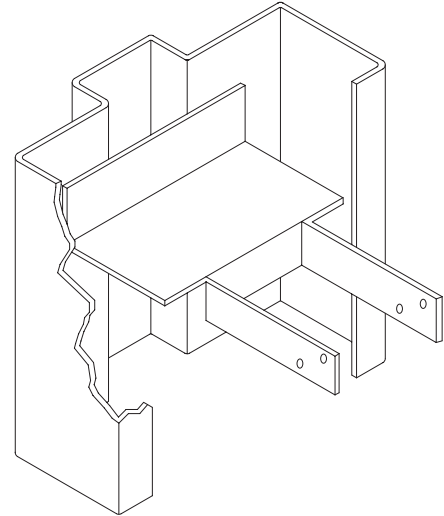


FIGURE 15C
COMBINATION WOOD/STEEL
STUD ANCHOR

Combination wood/steel stud anchor, Figure 15c, are welded or friction fit inside the jamb. Additional horizontal straps allows fastening to the face of the stud in lieu of through the throat. Typically used when the steel stud wall is assembled prior to setting the frame.

When fastening through the face of the stud, it is extremely important that the head height of fasteners are considered to ensure that the combined thickness of wall stud, fastener head height and finished wall material does not exceed the designed wall thickness. The effects of increased wall thickness will result in the frame faces being forced out during drywall installation, bowing the jamb profile. The corner between header and jamb will be visually bellied, and possible misalignment of frame may occur.

Due to this issue and with irregularities in wall construction width, it is strongly recommended that frames overlapping the wall, (except slip-on construction), have a throat dimension 1/8 in. (3.1mm) greater than dimensioned wall thickness.

Position the vertical steel studs in the frame throat opening in accordance with architect's details. Attach the vertical steel studs to floor and ceiling runners and fasten to the steel stud anchors with mechanical fasteners, Figure 15d.

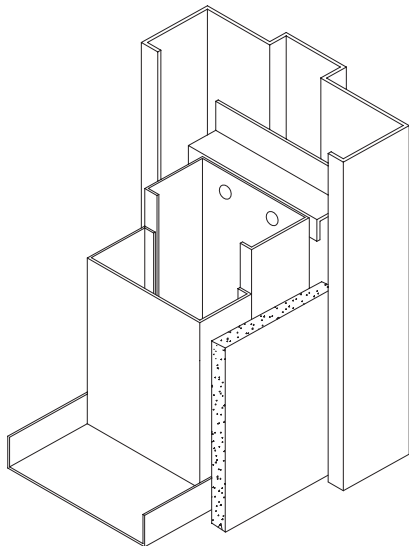


FIGURE 15D
STEEL STUD WALL WITH ANCHOR

Wall Construction: It is extremely important that the steel stud manufacturer's recommendation on thickness and general construction technique be followed to ensure that a solid and stable opening is achieved.

Figure 16a-d represents four methods of steel stud header connections to cripple studs above frame openings. Extreme precaution must be taken to ensure that any fasteners applied through face of studs or any construction will not increase the wall thickness.

Continually check squareness, plumbness, alignment, and twist in the frame as wall progresses.

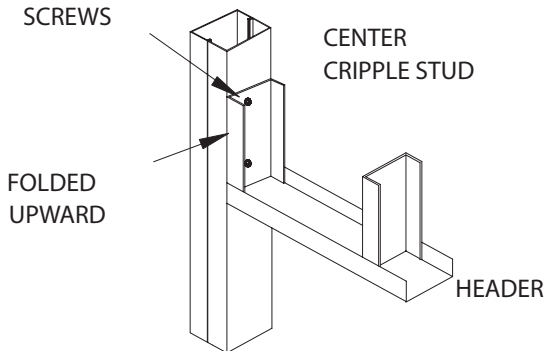


FIGURE 16A
RECOMMENDED

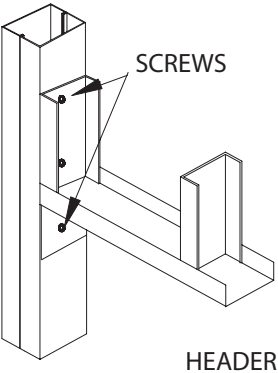


FIGURE 16B
ACCEPTABLE

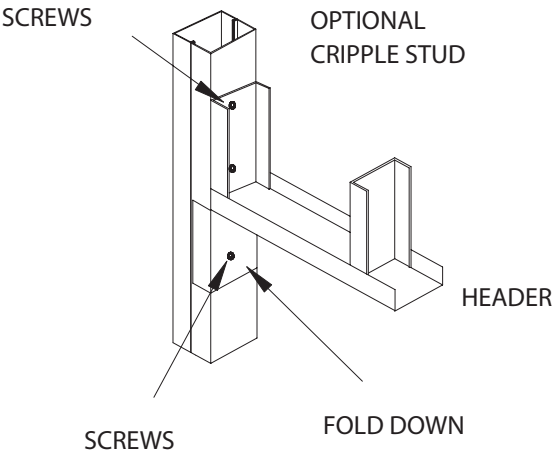


FIGURE 16C
NOT RECOMMENDED

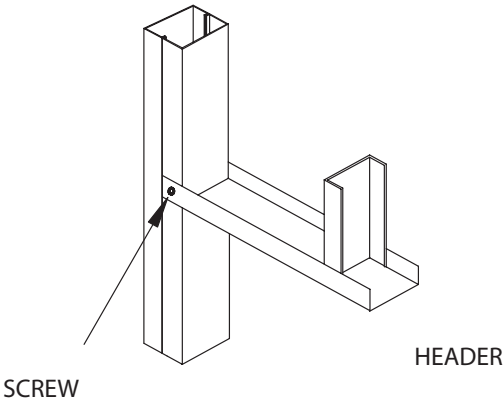


FIGURE 16D
NOT RECOMMENDED

3. WOOD STUD WALLS

Installation of a hollow metal frame in a wood stud wall, Figure 17a. Refer to HMMA 820 Hollow Metal Frames for additional anchorage methods.

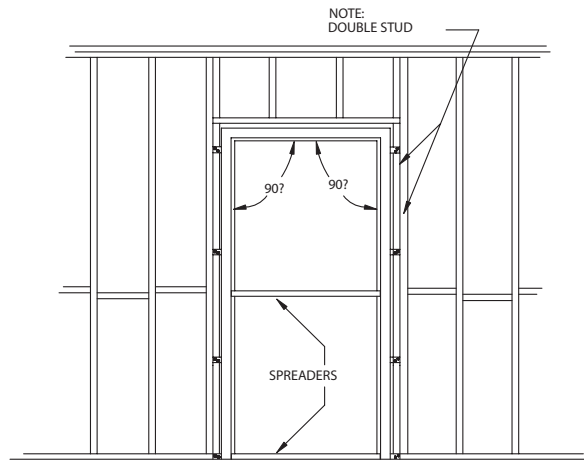


FIGURE 17A
WOOD STUD WALL

Wood stud walls can be constructed after the frame is set or prior to setting the frame. For constructing wall after the frame is set, follow guidelines same as steel stud walls. For constructing wall prior to setting the frame, follow these guidelines;

Rough Stud Opening:

- A. The width of the opening shall be the overall frame width plus 1/2 in. (12.7 mm).
- B. The height of the opening shall be the overall frame height plus 1/4 in. (6.4 mm).

Place the frame in the rough stud opening. Bend the anchor tabs around the stud, leaving the desired clearance between the frame return and stud, for inserting the finished wall material. Set the spreaders and level, plumb and align the frame. Check for twist in jamb. Shim by other and/or adjust as necessary

Square the frame at the top corner and nail the top anchor to the stud on ONE JAMB ONLY. Recheck the level, plumb and alignment of the frame at the other corner and continue to nail the balance of the anchors to the studs. Repeat the same process for the opposite jamb.

It is extremely important that the head of the fasteners are considered to ensure that the combined thickness of wall stud, fastener head height and finished wall material does not exceed the designed wall thickness.

Wood stud anchors, Figure 17b and 17c are welded to the back of the jambs. Figure 17d is either welded to back of the jamb or friction fit and available for wood or steel stud walls.

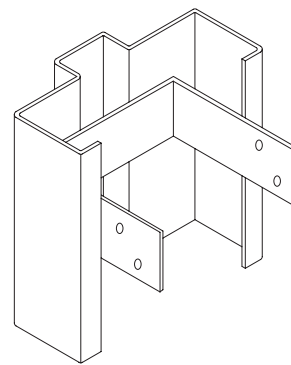


FIGURE 17B
WOOD-STUD ANCHORS

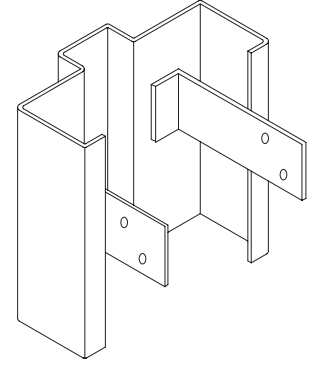


FIGURE 17C
WOOD-STUD ANCHORS

FIGURE 17D
COMBINATION WOOD/STEEL
STUD ANCHOR

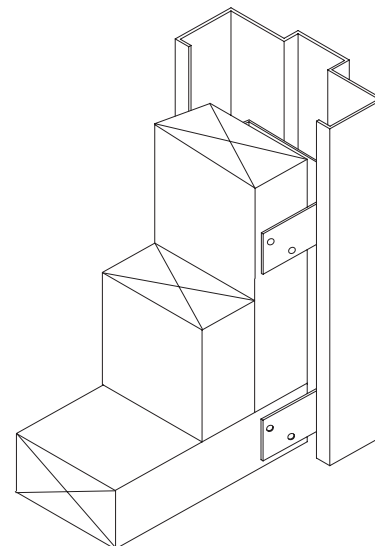


FIGURE 17E
WOOD STUD WALL WITH ANCHORS

4. EXISTING MASONRY WALLS

Refer to HMMA 820 Hollow Metal Frames for additional anchorage methods.

Clearance between existing wall and frame is critical in order to properly install frame. In addition to measuring the shortest dimension both vertically and horizontally, the existing opening must be checked for level and plumb. Size frame accordingly to dimensions taken providing 1/4 in. (6.3mm) clearance around frame.

Wall Preparation: Place the frame in the completed wall opening, Figure 18a. Check for squareness. Mark the wall through the holes in the jamb provided at the anchor points. Drill appropriate holes at the marks.

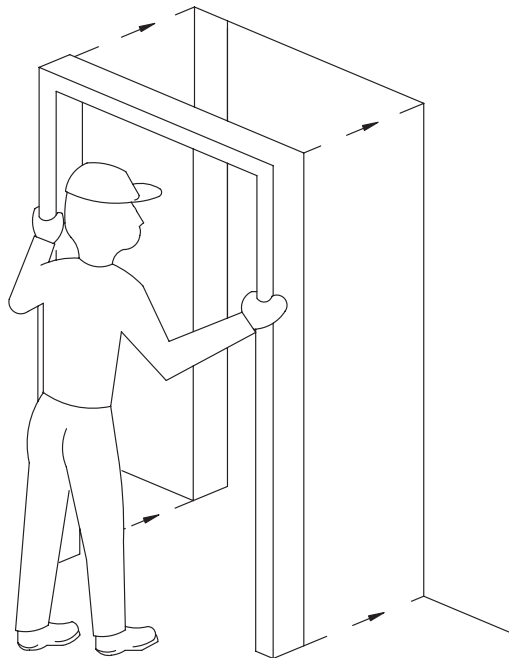


FIGURE 18A
EXISTING MASONRY WALL FRAME
INSTALLATION

Install frame: Insert mechanical fasteners through frame soffit into existing wall, Figure 18b. Shim, by others, behind anchors, above bolts to ensure a plumb frame.

The methods of plumbing the frame, and the use of the spreader noted under installation also applies to this wall condition. Tighten the screws. Caulk between frame and wall.

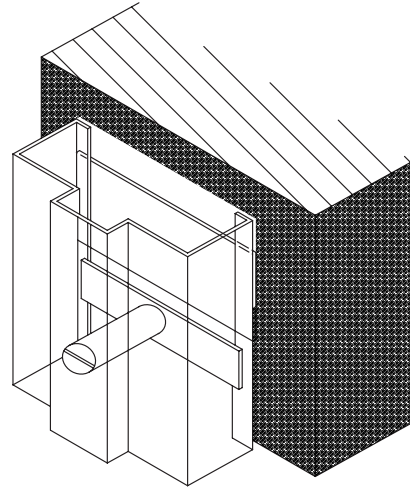


FIGURE 18B
EXPANSION BOLT INSTALLATION

Rough Buck Frame: Employ similar steps as described above when a Rough Buck is used, Figure 18c.

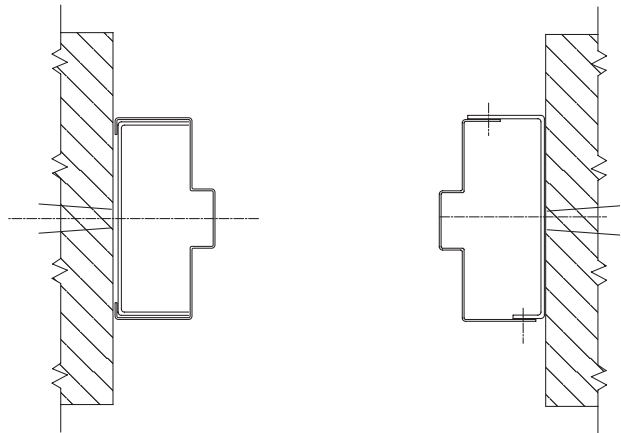


FIGURE 18C
ROUGH BUCKS

5. COMPLETED DRYWALL

Wall Construction: It is very important that the stud manufacturer's recommendation on thickness and general construction technique be followed to ensure that a solid and stable opening is achieved. For example, double studding at the opening is necessary; the header stud must be the same width as the jamb stud. It is particularly important that the overlapping of vertical and horizontal steel studs be avoided since this produces oversize walls. This could create significant installation problems when slip-on drywall frames are used.

Rough Stud Openings: A variety of anchoring techniques preclude the establishment of a definite rough opening standard. Follow the frame manufacturer's recommendations.

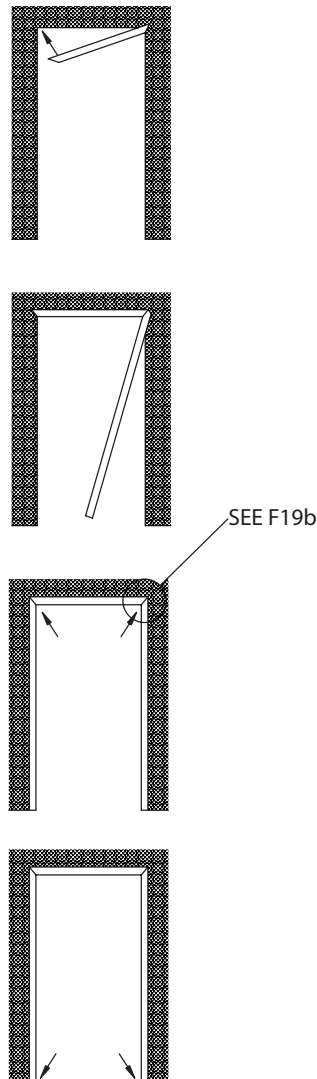


FIGURE 19A
INSTALLATION SEQUENCE

Installation Sequence: Installation varies from manufacturer to manufacturer. Consult the supplying manufacturer's literature for exact instructions. A typical installation cycle may be as follows, Figure 19a.

1. Slide the header in place over the wall approximately in the center of the opening.
2. Install one jamb by sliding it over the wall at the top. Push the bottom of the member until it is approximately vertical.
3. Install the other jamb in a similar manner. The frame should be installed with all excessive rough opening clearances thrown to the hinge side. The weight of the door will tend to cause movement, within the opening, toward the strike jamb.
4. Join the horizontal header to the jambs., Figure 19b . This is normally done by inserting tabs into the slots, or screws into the holes or a combination of both.

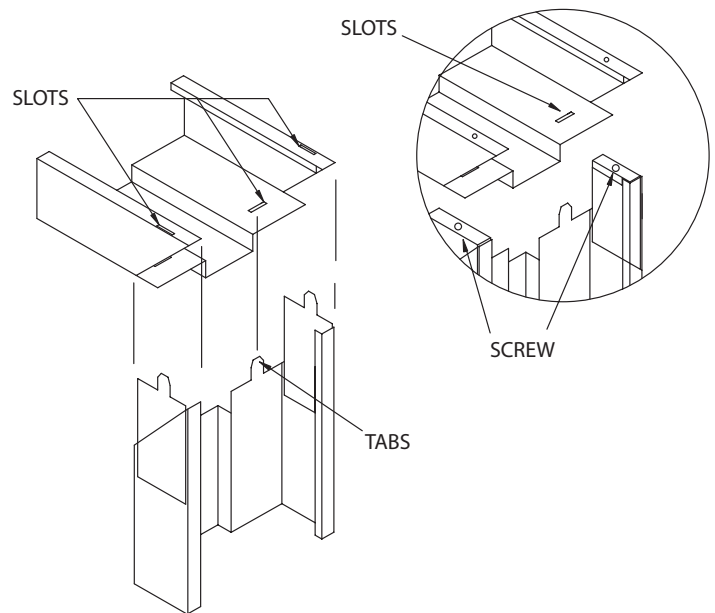


FIGURE 19B
CORNER CONNECTION
KNOCKDOWN SLIP-ON

Plumb and square the opening and check the jambs for twist. Some installers prefer to hang the door at this point, permitting the door to aid in performing this function.

Attach the base anchor to the stud or floor channel at the base of the wall.

The base anchor is usually a strap of metal, provided either welded or loose at each side of the jamb at the bottom with holes punched for nails or screws, Figure 19c.

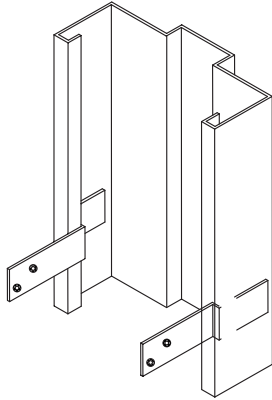


FIGURE 19C
SLIP-ON DRY WALL BASE ANCHOR

Compression Anchor, Figure 19d. An adjustable compression device is normally located near the top of each jamb. This anchor is used on slip-on dry wall frames and in conjunction with slip-on dry wall base anchors. Consult manufacturer's literature for direction to turn compression anchor. The compression anchor shall be hand tightened, do not use a screw gun.

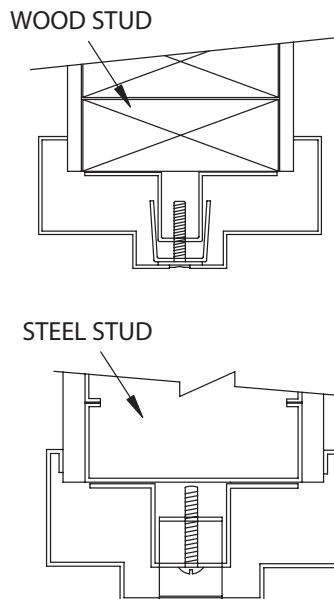


FIGURE 19D
COMPRESSION ANCHORS

F. GROUTING OF FRAMES

Grout guards and junction boxes are intended to protect hardware preparations and tapped mounting holes from masonry grout of 4 in. (100 mm) max. slump consistency which is hand troweled in place. If a lighter consistency of grout (greater than 4 in. (127 mm) slump when tested in accordance with ASTM C143) is to be used, special precautions must be taken in the field by the installation contractor to protect the hardware preparations. Any grout or other bonding material shall be promptly cleaned off of frames or doors following installation. Hollow metal surfaces shall be kept free of grout, tar or other bonding materials or sealers. Grouting materials which require air to dry (cure) are not recommended for such use in any closed section, such as a mullion.

When frames intended to be built into masonry construction which are prepared for silencers, the silencer must be installed prior to grouting to avoid filling the silencer hole.

See Appendix HMMA -820 TN01-03 "GROUTING HOLLOW METAL FRAMES"

G. FIELD SPLICING

When shipping and/or handling limitations so dictate, frames for large openings shall be fabricated in sections designed for splicing and welding in the field by others, Figure 20.

Field splices at labeled frames require either a mechanical or welded connection at each splice joint. Mechanical fasteners are through face of frame. If splice joints are welded, welds shall be ground smooth and immediately cleaned and painted with a compatible, direct to metal (DTM), rust inhibitive primer. For products fabricated from zinc coated steel, use zinc rich rust inhibitive primer.

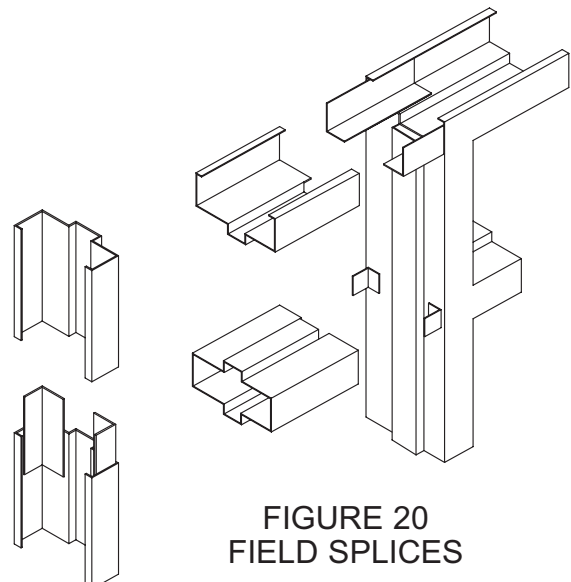


FIGURE 20
FIELD SPLICES

PART 4 - HANGING OF DOORS

A. GENERAL

It is the responsibility of the installer to hang all doors and install all hardware prior to finish painting. Doors shall be reinforced, drilled and tapped at the factory for templated mortise hardware only, in accordance with the approved hardware schedule and templates provided by the hardware supplier. Where surface mounted hardware, anchor hinges, thrust pivot, pivot reinforced hinges, or non-templated hardware apply, doors shall be reinforced, with drilling and tapping done in the field by others.

Experienced craftsmanship and care are essential in the hanging of metal doors. The use of steel hinge shims may be required to provide uniform clearance around the door and alleviate "hinge bind". Figure 21 and Figure 22.

The door-to-frame clearance adjustment toward the strike jamb (in the direction of arrow S) in Figure 21. can be accomplished by placing a metal shim or shims under the hinge leaf or leaves along the barrel edge of the hinge. Adjusting toward the hinge jamb (in the direction of arrow H) in Figure 22 can be done by placing a metal shim or shims under the outer edge of the hinge leaf or leaves.

B. OPERATION CLEARANCES

Edge clearances for swinging hollow metal doors and as specified in NFPA 80, shall be provided for the functional operation of the assembly and shall not exceed the following (for all door heights)

Between door and frame at head and jamb. 1/8 in. (3.1 mm) +/- 1/16 in. (1.5 mm)

Between edges of pairs of doors. 1/8 in. (3.1 mm) +/- 1/16 in. (1.5 mm)

Floor clearance for fire rated swinging hollow metal doors shall not exceed 3/4 in. (19mm). Floor clearance shall be provided for the functional operation of all swinging hollow metal doors and shall not be less than 1/8 in. (3.1mm)

The Architect must define the distance from the top of the floor / finished floor to top of floor covering so appropriate undercuts can be provided. Floor / Finish Floor is defined as the top of the concrete or structural slab. HMMA uses the term "top of floor covering" to describe the NFPA term "nominal surface of floor covering". Please refer to HMMA-810 TN01-03 Tech Note, "Defining Undercuts."

C. CARE AFTER INSTALLATION

Doors installed prior to completion of construction may be subject to damage from other trades because of improper protection or carelessness on the part of construction workers.

Prime or painted surfaces which have been scratched or otherwise marred during installation (including field welding) and/or cleaning, shall promptly be finished smooth, cleaned, treated for maximum paint adhesion, and touched up with a direct to metal (DTM), rust inhibitive primer.

Acid washing of masonry construction adjacent to door and frame installations will damage the steel surfaces whether primed or finish painted.

If not cleaned immediately, rusting will occur. For this reason steel doors and frames must be protected. If exposed to an acid wash, all primed or painted steel surfaces must be thoroughly cleaned with particular attention being given to hidden areas such as those areas under glazing stops.

For additional information regarding the painting of hollow metal products, please refer to HMMA-840 TN01-07 Tech Note, "Painting Hollow Metal Products"

USE HINGE SHIMS TO ADJUST DOOR CLEARANCE

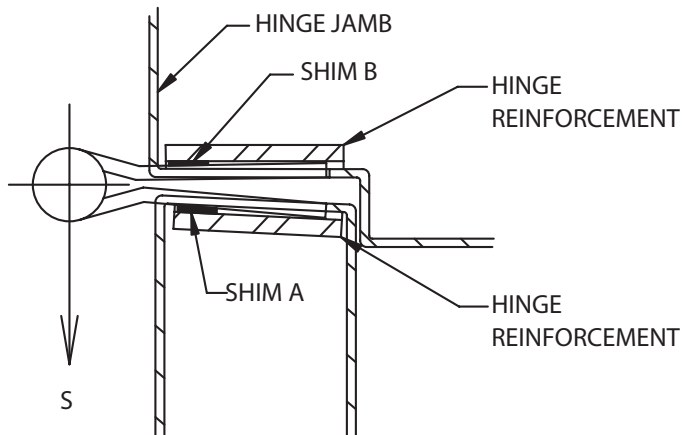


FIGURE 21
SHIMMING TO INCREASE
CLEARANCE AT HINGE EDGE

Using shim A only, door will be relocated in direction of arrow S.

Using shim B only, both door and centerline of hinge barrel will move in direction of arrow S.

Using both shims A and B will move the door further in direction of arrow S than by using either A or B alone, and hinge barrel will be relocated just as using B alone.

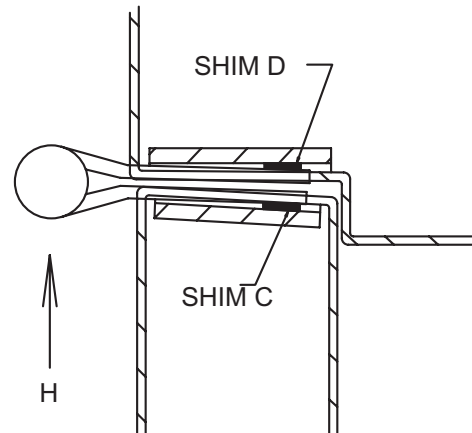


FIGURE 22
SHIMMING TO DECREASE
CLEARANCE AT HINGE EDGE

Using shim C only, door will be relocated in direction of arrow H.

Using shim D only, Both door and centerline of hinge barrel will move in direction of arrow H.

Using both shims C and D will move the door further in direction of arrow H than by using either C or D alone, and hinge barrel will be relocated just as by using D alone.